
StepCAD: Mesh-to-CAD Code Generation via LLM Policy and Geometry-Guided Search

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Abstract

Recovering executable CAD programs from 3D meshes is challenging due to the compositional nature of CAD construction and the interaction between discrete modeling choices and continuous parameters. Many learning-based methods predict complete programs in a single pass and rely predominantly on sketch-extrude representations, limiting operation diversity and opportunities to correct geometric errors during reconstruction. We introduce StepCAD, a generative optimization approach that combines a state-conditioned CAD policy with geometry-guided search. Given an input mesh, the policy predicts construction actions conditioned on both target and intermediate geometry, and an IoU-guided tree search refines the resulting program through local edits. We also introduce ARCADE-1.5M, a large-scale dataset of 1.5M executable CAD programs spanning diverse operations, sequences with a maximum length of 150+ counted operations, and 12.5M intermediate state-action transitions. Experiments across multiple CAD reconstruction benchmarks show that StepCAD achieves state-of-the-art geometric reconstruction accuracy with consistently high validity, yielding up to 87.2% relative IoU improvement over the strongest evaluated baseline, with particularly large gains on complex shapes. Project page: <https://ghadinehme.com/stepcad.github.io/>

1 Introduction

Recovering structured, executable CAD programs from 3D meshes is a critical capability for modern engineering workflows. Many industrial pipelines produce dense, unstructured 3D geometry — from 3D scans of physical parts [1, 2], topology-optimized designs [3, 4], and increasingly from generative 3D models [5, 6, 7] — that must be converted into editable parametric representations to enable design iteration, manufacturing, and downstream simulation. Our goal is geometry-driven reverse engineering: recovering an executable construction sequence that reproduces the input, without claiming recovery of its original design intent or constraints. Unlike conventional 3D reconstruction, which approximates surface geometry, *mesh-to-CAD generation* seeks to recover a sequence of parametric operations that reconstructs the shape compactly and faithfully. This task is inherently challenging: CAD programs are long-horizon, compositional, and highly structured, with strong dependencies between operations and continuous parameters that must be estimated with high precision [8, 9].

Recent learning-based approaches model CAD reconstruction as autoregressive program generation conditioned on geometry [8, 2, 10, 11]. Early approaches and datasets emphasize sketch-and-extrude pipelines, while recent procedural datasets expand the use of standard CAD operations [12, 13]. Nevertheless, predicting a complete program without executing intermediate states limits opportunities to respond to the evolving reconstruction. Learned refinement methods introduce feedback, but geometric errors can remain in both the program structure and its continuous parameters.

This motivates combining state-conditioned generation with explicit, executable program-space search.

We argue that mesh-to-CAD reconstruction should be reformulated as policy-driven, geometry-conditioned program synthesis with search-based refinement. This perspective decomposes the problem into two complementary components: a learned policy that conditions on intermediate geometry to propose candidate construction sequences, and a geometry-aware optimization procedure that refines these sequences to improve geometric fidelity. The formulation leverages the strengths of generative models for capturing global structure while correcting geometric discrepancies through explicit search guided by geometric objectives — a paradigm with a long history in program synthesis [14, 15, 16] that we adapt to the unique constraints of CAD generation.

Realizing this paradigm requires both a model capable of state-conditioned generation and training data that exposes diverse construction trajectories — neither of which is well-supported by existing CAD datasets. We therefore introduce two contributions in tandem. First, ARCADE-1.5M, a large-scale synthetic dataset of 1.5M executable CAD programs spanning extrude, revolve, loft, sweep, Boolean operations, and finishing features, with sequences reaching 150+ counted operations and 12.5M intermediate state-action transitions providing dense supervision over construction trajectories. Second, StepCAD, a generative optimization [17] approach for mesh-to-CAD reconstruction that, given an input shape, generates a candidate program using a state-conditioned LLM policy trained on ARCADE-1.5M, then refines it through a geometry-aware tree search over local edits — combining continuous parameter optimization with discrete structural modifications, all guided by IoU.

This combination of policy and optimization yields substantial improvements in reconstruction accuracy and robustness. We show that explicit refinement of program structure and numerical parameters improves geometric fidelity beyond generation alone. Here, geometric fidelity refers to agreement with the target shape, not recovery of every parameter in a reference program. StepCAD achieves state-of-the-art (SOTA) performance on CADBench [18], a CAD benchmark spanning ABC [19], Fusion 360 [20], DeepCAD [8], MCB [21], and Objaverse [22], with increasingly larger gains as shape complexity grows, precisely where prior approaches degrade most severely.

In summary, our contributions are:

- **ARCADE-1.5M: a large-scale CAD program dataset.** We introduce a dataset of 1.5M executable CAD programs with rich operation diversity, long-horizon construction sequences (150+ counted operations), and 12.5M intermediate state-action transitions, substantially expanding the scale and complexity of available CAD data.
- **State-conditioned CAD policy.** We propose a point-cloud-conditioned autoregressive policy that predicts CAD operations given both intermediate and target geometry.
- **Program-space optimization for CAD generation.** We reformulate mesh-to-CAD reconstruction as optimization over program space and introduce a geometry-aware tree search that refines candidate programs through IoU-guided local edits.
- **State-of-the-art performance on diverse CAD benchmarks.** We achieve SOTA results across five diverse benchmarks, with up to **87.2%** relative improvement in IoU over the best prior method, demonstrating strong robustness and growing advantages as shapes become more complex.

2 Related Work

Learning-Based CAD Program Generation. While text-to-CAD generation has attracted significant attention [23, 24, 25], industrial workflows more often require recovering CAD programs from existing 3D geometry — such as scanned parts [1, 2, 26], topology-optimized geometry [3, 4, 27], or AI-generated meshes [5, 6, 7] — where the design already exists in shape but lacks editable parametric structure. Recent work has therefore explored neural approaches for generating CAD programs as sequences of parametric operations from 3D shapes [2, 9, 10], motivated by applications in reverse engineering, scalable dataset construction, and bridging geometry-centric 3D generation [5, 6] with editable, parametric workflows. Early methods such as DeepCAD [8] introduce domain-specific languages for sketch-and-extrude programs and learn sequence autoencoders over this representation, with subsequent works extending the paradigm to richer inputs and structured decoders [28, 29, 9, 30]. More recently, large language models have been applied to CAD program generation from images [7, 31, 32] and point clouds [2, 10, 11], enabling flexible conditioning across modalities. Despite these

advances, existing methods rely primarily on amortized autoregressive prediction, which can leave geometric discrepancies in the resulting shape. In contrast, StepCAD explicitly separates program generation from program optimization, using a learned policy to propose structurally consistent programs and a search procedure to refine them for geometric accuracy.

Iterative Refinement and Neural-Guided Program Search. A small but growing line of work explores iterative refinement for CAD generation. CADReasoner [33] introduces LLM-based refinement guided by visual feedback, CADCrafter [31] applies direct preference optimization with a code-validity checker, and CADFusion [24] alternates supervised and visual-feedback stages during training. These methods incorporate feedback through different training or generation procedures. StepCAD instead makes execution-scored local program edits an explicit inference-time optimization stage. More broadly, combining learned policies with explicit search has a long history in program synthesis [14, 15, 16] and recent code generation [34, 35, 36], where neural networks propose candidates that are refined by search to satisfy correctness criteria. StepCAD adopts this paradigm in the CAD setting: an LLM policy proposes structurally consistent programs, while a geometry-aware tree search refines them through local edits — including continuous parameter optimization, sketch modification, and operation skipping — with candidates ranked by IoU against the target. Compared to general code generation, CAD program search is uniquely constrained by both executability and continuous parameter precision, motivating geometry as the primary search signal.

Entity-grounded and feedback-driven CAD generation. Pointer-CAD [37] conditions on text and intermediate B-reps and selects referenced edges and faces through a pointer mechanism. FutureCAD [38] grounds textual selection queries to B-rep primitives. Seek-CAD [39] uses visual and chain-of-thought feedback for iterative generation. Our focus is mesh-conditioned reconstruction, where an explicit target shape enables direct geometric scoring of executable program edits. StepCAD combines this objective with state-conditioned proposals and a refinement stage that can also process compatible programs from other generators. Its coarse selectors do not provide the fine-grained entity addressing of Pointer-CAD or FutureCAD. Contemporaneous Pointer-CAD v2 [40] further motivates distinguishing geometric overlap from dimension-aware and entity-level accuracy.

Geometry-Driven CAD Reverse Engineering. Another line of work recovers CAD structure directly from 3D shape [41]. Classical pipelines decompose shapes into primitives and fit analytic surfaces but struggle to produce consistent parametric programs for complex designs. Learning-based approaches improve robustness by combining neural segmentation with surface fitting, particularly for boundary representation (B-rep) reconstruction [26, 42, 43, 44, 1]. B-rep reconstruction alone does not recover an ordered feature history; recovering geometry or an executable history also does not establish recovery of design intent. Inverse constructive solid geometry (CSG) methods [45] formulate reconstruction as a combinatorial search over Boolean programs, but are typically restricted to simple primitive sets and do not scale to sketch-based modeling, revolutions, or finishing features. In contrast, StepCAD explicitly models the sequential construction process and produces executable programs, enabling both interpretability and editability.

CAD Datasets. Large-scale datasets such as ABC [19], DeepCAD [8], and Fusion360 [20] have enabled learning structured CAD representations. Their supervision differs: DeepCAD and the Fusion 360 Gallery construction task emphasize sketch-and-extrude sequences, whereas ABC supplies geometry without an ordered construction history. Concurrent efforts such as CAD-Recode [2], Zero-to-CAD [12], and CADEvolve [13] expand operation diversity through procedural and evolutionary synthesis, using many of the same standard CadQuery operations as our corpus. Our ARCADE-1.5M dataset is designed to provide long-horizon, diverse, and executable CAD programs with dense intermediate supervision, enabling effective training of state-conditioned policies.

3 Method

We hypothesize that learning CAD programs from geometry requires large-scale procedural data with rich operation diversity, which is lacking in existing datasets. To address this, we introduce **ARCADE (Artificial Repository of CAD Examples)**, a large-scale synthetic dataset of executable CAD programs with diverse operations and workflow-consistent structure, providing dense supervision over construction trajectories. We then propose **StepCAD**, a generative optimization approach for mesh-to-CAD reconstruction: a **state-conditioned LLM policy** generates candidate programs,

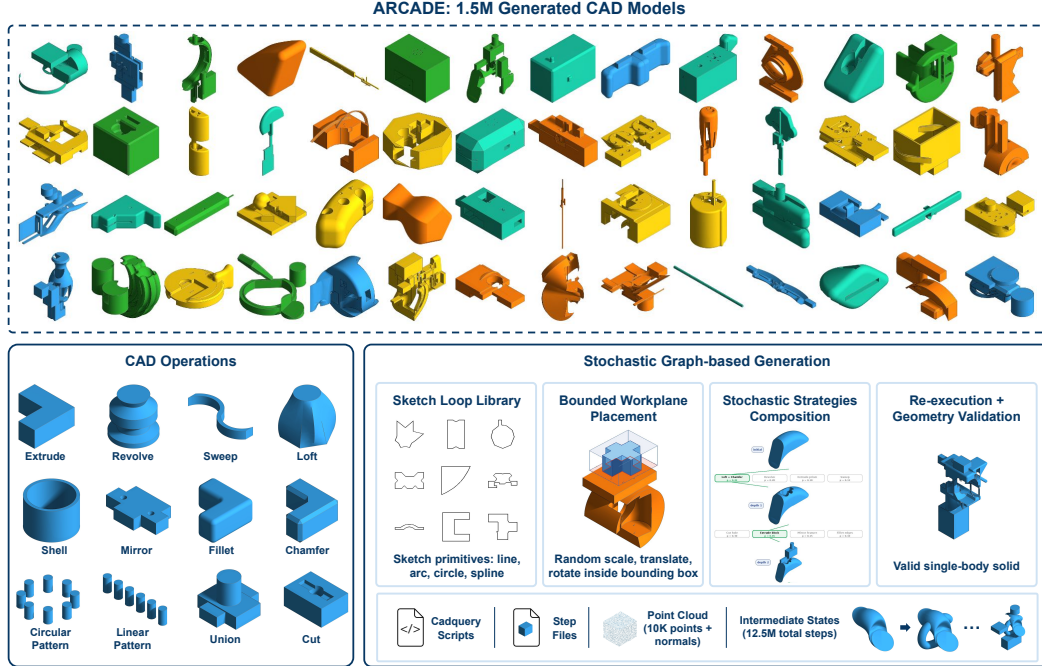


Figure 1: **ARCADE-1.5M: large-scale synthetic CAD program generation.** Our pipeline synthesizes diverse long-horizon CAD programs through stochastic strategy-guided composition of sketch primitives and CAD operations. Each program is executed and validated, yielding 1.5M executable programs and 12.5M intermediate construction states with broad operation diversity.

followed by **geometry-aware optimization** in program space. This decomposition enables learning global construction structure while achieving high geometric fidelity.

3.1 Problem Statement

We study **mesh-to-CAD reconstruction**: recovering an executable parametric CAD program that reproduces a target 3D shape. Let $\mathcal{M}$ denote the space of meshes and $\mathcal{P}$ the space of valid CAD programs. A program $p = (a_1, \dots, a_T) \in \mathcal{P}$ is an ordered sequence of construction actions whose execution produces geometry $S(p) \in \mathcal{M}$. Given an input mesh $m \in \mathcal{M}$, the goal is to predict a valid program $\hat{p}$ such that the generated geometry matches the target shape: $\hat{p} = \arg \max_{p \in \mathcal{P}} \text{IoU}(S(p), m)$. The problem is challenging due to the combinatorial structure of $\mathcal{P}$, long-horizon autoregressive dependencies, and the existence of multiple valid programs producing similar geometry. We represent CAD programs using CadQuery code [46].

3.2 ARCADE-1.5M: Constraint-Guided Synthetic CAD Programs

We introduce **ARCADE-1.5M**, a dataset of **1.5M** executable parametric CAD programs paired with valid 3D geometry. ARCADE-1.5M combines stochastic geometry generation with workflow-inspired constraints to synthesize long-horizon, structurally plausible CAD construction sequences at scale (Fig. 1). Dataset statistics and generation details can be found in Appendices G and H.

Constraint-guided primitive generation. We initialize generation from a library of 2D sketch loops extracted from ABC, including lines, arcs, circles, and splines. Sketches are randomly perturbed, rotated, and scaled, then placed on randomly selected faces of a target bounding box, providing explicit control over feature size, placement, and orientation. Each sketch is converted into a solid using randomly sampled **extrude**, **revolve**, **loft**, or **sweep** operations with stochastic parameters. Optional fillets and chamfers are applied probabilistically to increase geometric diversity.

Workflow-inspired composition constraints. Rather than composing disconnected primitives, ARCADE-1.5M enforces constraints inspired by practical CAD workflows. All intermediate states

Dataset	Size	Has Construction Sequence	Avg Ops / Seq	Avg Solid Ops / Seq	Avg Faces	Solid Operation Coverage (%)							
						Extrude	Revolve	Sweep	Loft	Fillet	Chamfer	Shell	Mirror
DeepCAD [8]	178k	✓	15	1.5	10	100	0	0	0	0	0	0	0
ABC-mono [47]	200k	✓	*	*	*	100	0	0	0	0	0	0	0
KOCAD [47]	4.6k	✓	*	*	*	100	0	0	0	0	0	0	0
Fusion 360 [20]	8.6k	✓	18	2.2	16	100	0	0	0	0	0	0	0
CAD-Recode [2]	1M	✓	28	2.0	17	100	0	0	0	0	0	0	0
Zero-to-CAD [12]	1M	✓	24	5.5	44	85.99	18.73	3.88	4.17	36.92	74.66	15.00	1.56
CADEvolve [13]	1.3M	✓	28	5.5	54	83.05	4.80	5.75	8.48	27.78	4.76	1.95	0.08
ABC [19]	1M	✗	—	—	114	—	—	—	—	—	—	—	—
ARCADE-1.5M (ours)	1.5M	✓	40	8.5	86	95.42	12.13	25.37	13.10	34.77	33.66	10.01	6.70

Table 1: **Comparison of CAD datasets with operation coverage.** Operation coverage is the percentage of programs containing at least one instance of the operation. Prior datasets are mostly sketch-extrude only. Concurrent works CADEvolve and Zero-to-CAD add parametric diversity, while ARCADE-1.5M has the most programs, highest mean operation counts, and highest mean face count among the listed datasets with construction sequences. *: datasets that are not publicly available.

must remain valid single-body solids with positive volume. Additive operations require controlled overlap with the host body, while subtractive operations are accepted only under sufficient volumetric intersection. These constraints prevent degenerate geometry and preserve physically meaningful feature placement.

Strategy-based hierarchical construction. To generate long, coherent programs, ARCADE-1.5M uses **94 heuristic generation strategies** that constrain operation types and approximate sequence lengths. Strategies emulate common modeling patterns, including additive construction, cut-heavy machining, shelling, mirroring, lofting, sweeping, and finishing operations. This hierarchical structure enables realistic long-horizon programs while avoiding combinatorial explosion. Generated sequences range from 3 to 150+ operations (mean: 40.3, median: 33).

Operations, actions, and API calls. An *operation* in our dataset statistics is a sketch primitive, solid-modeling operation, or Boolean combination. A *solid operation* is an extrude, revolve, sweep, loft, fillet, chamfer, shell, or mirror. A policy *action* bundles a solid operation with its associated Boolean combination and, for sketch-based operations, the profile-defining sketch primitives. Finishing operations do not require a new sketch. Thus, the mean of 40.3 counted operations per program differs from the mean of 8.5 solid operations (median 8), corresponding to approximately $40.3/8.5 = 4.7$ counted operations per construction action. An API call is a code-level method invocation; setup, selection, and export calls should not be confused with the operation-count categories. Actions are defined at geometry-update boundaries so that the next policy step can condition on a new solid.

Validation and normalization. Every sample undergoes double validation: the in-memory CadQuery solid must be valid, and the exported Python program is re-executed in a clean namespace before STEP export. Shapes are normalized to a canonical $[-1, 1]^3$ bounding box, and floating-point literals are rounded to three decimals to reduce token complexity.

Dataset characteristics. Each sample contains a CadQuery script, STEP B-Rep, triangulated mesh, and a 10K-point cloud with normals. We additionally store intermediate geometry after every **solid operation**, yielding approximately **12.5M** construction states for dense trajectory supervision. ARCADE-1.5M covers workplanes, sketch primitives, volumetric and Boolean operations, shells, mirrors, fillets, chamfers, and spatial patterns. Its contribution is the distribution and composition of these standard operations, together with scale and intermediate supervision, rather than a new operation vocabulary. For example, sweep occurs in 25.37% of programs, compared with 3.88% for Zero-to-CAD and 5.75% for CADEvolve; the mean face count is 86 (Table 1). Overall, ARCADE-1.5M bridges executable CAD construction logic and resulting 3D geometry, enabling training for realistic mesh-to-CAD generation.

3.3 StepCAD: Generative Optimization for Mesh-to-CAD Reconstruction

We propose **StepCAD**, a two-stage framework that combines a *state-conditioned autoregressive CAD policy* (II-StepCAD) with *geometry-aware program optimization* (Fig. 2). The policy generates a structurally plausible executable program, while a subsequent local search stage refines the program to improve geometric fidelity.

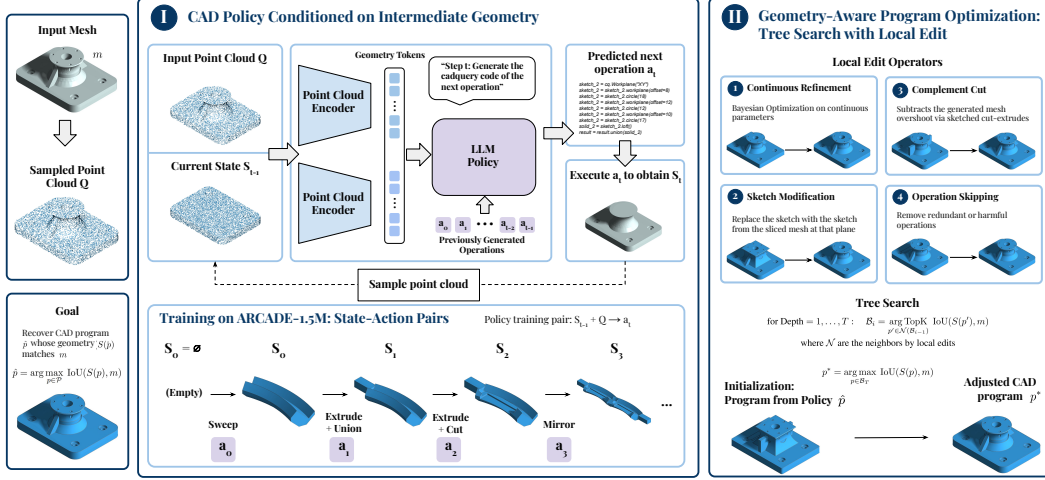


Figure 2: **StepCAD pipeline.** A state-conditioned CAD policy generates an executable program from an input shape, which is subsequently refined via geometry-aware search over local program edits.

3.3.1 II-StepCAD: State-Conditioned CAD Policy

Sequential generation. Direct optimization in program space is combinatorial and highly non-convex. We instead learn an autoregressive policy

$$\pi_{\theta}(a_t \mid Q_{t-1}, Q_m, a_{<t}),$$

where Q_m is a point cloud sampled from the target mesh, Q_{t-1} is sampled from the intermediate geometry, and $a_{<t}$ denotes previously generated construction actions. This induces a distribution over executable CAD programs:

$$\pi_{\theta}(p) = \prod_{t=1}^T \pi_{\theta}(a_t \mid Q_{t-1}, Q_m, a_{<t}).$$

Geometry conditioning. Target and intermediate geometries are encoded using a shared Point-MAE encoder [48]:

$$z_m = f_{\phi}(Q_m), \quad z_{t-1} = f_{\phi}(Q_{t-1}),$$

and injected as conditioning tokens into the autoregressive model.

Policy architecture. We instantiate π_{θ} using Qwen2.5-Coder-1.5B [49] conditioned on geometry embeddings and prior program tokens. The model predicts executable construction actions directly in token space.

Training. The policy is trained with teacher forcing over dense state-action trajectories extracted from ARCADE-1.5M:

$$\mathcal{L}(\theta) = -\mathbb{E}[\log \pi_{\theta}(a_t \mid Q_{t-1}, Q_m, a_{<t})].$$

Supervision at every intermediate construction step improves long-horizon stability and geometric consistency.

Closed-loop inference. At inference, generation alternates between policy prediction and CAD execution. Starting from an empty state, the model performs inference-time sampling to predict the next construction action, executes the updated CadQuery program, samples the resulting geometry, and reconditions on the produced intermediate shape before continuing generation. Generation terminates once reconstruction quality plateaus, an end token is emitted, or a cap of $T_{\max} = 12$ construction actions is reached (Details in Appendix I). This produces a seed program $\hat{p}$ that captures the global structure of the target shape and initializes the refinement stage.

3.3.2 Geometry-Aware Program Optimization

Given a seed CAD program $\hat{p} \sim \pi_{\theta}$, we perform local search in program space:

$$p^* = \arg \max_{p \in \mathcal{N}(\hat{p})} \text{IoU}(S(p), m),$$

where $S(p)$ denotes the executed geometry and $\mathcal{N}(\hat{p})$ is defined by local program edits.

Direct optimization over all edits is exponential in the program length T . We therefore perform a geometry-aware beam search over operations in execution order. At step t , each candidate program in the beam is expanded using a small set of local edits applied to operation a_t , executed in CadQuery, and ranked by IoU against the input mesh: $\mathcal{B}_t = \text{TopK}_K \{p' \in \text{Expand}(p, t) : p \in \mathcal{B}_{t-1}\}$.

We use beam width $K = 2$ and four local edits per expansion, giving a number of edit expansions linear in T for fixed K , edit count, and per-edit budget. The final refined program is selected as:

$$p^* = \arg \max_{p \in \mathcal{B}_T} \text{IoU}(S(p), m).$$

The local edits are:

- **Continuous refinement:** optimize continuous operation parameters (e.g., depths, radii, sketch dimensions).
- **Sketch modification:** replace operation sketches using geometry extracted from the input mesh.
- **Complement cut:** remove overshoot regions using automatically generated cut-extrudes.
- **Operation skipping:** remove redundant or harmful operations.

Implementation Details. All algorithmic and training details are provided in appendices I and J.

4 Experiments, Results, and Discussion

We evaluate **StepCAD** on multiple mesh-to-CAD reconstruction benchmarks, comparing against state-of-the-art methods and analyzing the contributions of each component of our approach. We focus on geometric fidelity, validity of generated programs, and robustness to complex constructions.

Benchmark. We evaluate on **CADBench** [18], a comprehensive benchmark for CAD program generation that aggregates five representative datasets spanning diverse geometric domains and construction complexity. Specifically, CADBench includes: **DeepCAD** [8] and **Fusion 360** [20], **ABC** [19] for large-scale CAD shapes with varying difficulty (Easy, Medium, Hard, based on number of STEP faces [50]), **MCB** [21] for mechanical components, and **Objaverse** [22] for organic shapes. Each benchmark contains 3,000 samples, except ABC which provides 2,000 samples per complexity split. This suite enables robust evaluation across both structured mechanical designs and free-form geometries, providing a holistic assessment of CAD generation performance.

Metrics. We report **Intersection-over-Union (IoU)** measuring volumetric overlap, **Chamfer Distance (CD)** measuring surface discrepancy (lower is better), and **Valid Shape Rate (VSR)**, the percentage of generated programs that execute successfully and yield valid geometry. (Appendix F)

Baselines. We compare against CAD-Recode [2], CADEvolve [13], and Cadrille [10] which are representative learning-based methods for CAD generation from geometry, as well as CADReasoner [33], an iterative CAD refinement for mesh to CAD generation. The main benchmark table evaluates publicly released checkpoints. Because these models use different training corpora, we additionally train a CAD-Recode-style single-shot Qwen2.5 baseline on ARCADE-1.5M (Table 3). This baseline predicts a complete program from the target point cloud without intermediate-state conditioning or refinement.

Quantitative Results. Table 2 reports IoU and Chamfer Distance on CADBench across structured, mechanical, and organic domains. StepCAD achieves SOTA performance on all benchmarks, with the largest gains on complex shapes.

State-of-the-art mesh-to-CAD reconstruction. StepCAD consistently outperforms prior methods across all datasets. On **ABC Hard**, IoU improves from 0.31–0.39 to **0.73**, an **87.2%** relative gain over the strongest baseline. On **ABC Medium**, IoU increases from 0.43 to **0.74 (72.1% gain)**. On the **Mechanical** and **Organic** subsets, StepCAD reaches **0.89** and **0.71** IoU, improving by **32.8%** and **69.0%** respectively. StepCAD also achieves the lowest Chamfer Distance on every benchmark, with reductions up to **28.0%** on Mechanical, demonstrating robustness across both structured CAD models and free-form geometries.

Table 2: **Mean mesh-to-CAD generation metrics.** IoU and VSR are higher better; CD is lower better. Best overall in bold; second best underlined.

Model	DeepCAD			Fusion 360			ABC Easy			ABC Medium			ABC Hard			Mechanical			Organic		
	IoU $\uparrow$	CD $\downarrow$	VSR $\uparrow$	IoU $\uparrow$	CD $\downarrow$	VSR $\uparrow$	IoU $\uparrow$	CD $\downarrow$	VSR $\uparrow$	IoU $\uparrow$	CD $\downarrow$	VSR $\uparrow$	IoU $\uparrow$	CD $\downarrow$	VSR $\uparrow$	IoU $\uparrow$	CD $\downarrow$	VSR $\uparrow$	IoU $\uparrow$	CD $\downarrow$	VSR $\uparrow$
CAD-Recode	0.79	0.0207	96.22	0.65	0.0268	92.63	0.76	0.0237	95.02	0.37	0.0349	88.35	0.35	0.0328	87.57	0.66	0.0297	94.01	0.40	0.0570	87.84
Cadrlle	0.78	0.0211	96.97	0.66	0.0268	94.87	0.75	0.0234	96.58	0.40	0.0359	92.30	0.37	0.0337	92.18	0.65	0.0295	96.80	0.42	0.0549	91.03
CADEvolve	0.74	0.0389	99.70	0.63	0.0466	97.60	0.72	0.0413	98.88	0.39	0.0638	95.82	0.39	0.0617	95.27	0.64	0.0486	96.90	0.40	0.0890	96.91
CADReasoner	<u>0.86</u>	<u>0.0197</u>	99.47	0.71	0.0326	<u>98.33</u>	<u>0.83</u>	0.0242	99.60	0.43	0.0419	97.10	0.31	0.0627	92.80	0.67	0.0413	98.80	0.31	0.0968	96.60
Π -StepCAD (Ours)	0.82	0.0205	100.00	<u>0.75</u>	<u>0.0240</u>	99.97	0.77	<u>0.0227</u>	100.00	<u>0.53</u>	<u>0.0320</u>	100.00	<u>0.52</u>	<u>0.0283</u>	99.95	<u>0.78</u>	<u>0.0259</u>	100.00	<u>0.53</u>	<u>0.0446</u>	100.00
StepCAD (Ours)	0.89	0.0191	100.00	0.83	0.0228	99.97	0.89	0.0200	100.00	0.74	0.0302	100.00	0.73	0.0263	99.95	0.89	0.0213	100.00	0.71	0.0420	100.00
Improvement vs. Best Prior (%)	3.5	3.0	0.3	16.9	14.9	1.7	7.2	14.5	0.4	72.1	13.5	3.0	87.2	19.8	4.9	32.8	27.8	1.2	69.0	23.5	3.2

Benefits of state-conditioned step-by-step generation. Even without optimization, Π -StepCAD already surpasses prior work, especially on difficult benchmarks (e.g., 0.52 IoU vs. ~ 0.38 on ABC Hard, $\sim 37\%$ improvement). Unlike one-shot autoregressive approaches, Π -StepCAD conditions each prediction on the intermediate geometry produced so far, enabling progressively more accurate structural and parametric decisions throughout construction. We note that CADReasoner performs slightly better than Π -StepCAD on the easiest benchmarks (DeepCAD and ABC Easy), likely due to specialization to simpler in-distribution samples. However, its performance degrades substantially on harder subsets, whereas Π -StepCAD remains robust. This suggests that explicit intermediate-geometry conditioning generalizes better to complex shapes than iterative learned refinement alone.

Generative optimization is critical. The full StepCAD system substantially improves over the policy alone (e.g., 0.52 $\rightarrow$ 0.73 IoU on ABC Hard). The geometry-aware tree search refines program structure and numerical parameters (e.g., extrusion depths and sketch dimensions), producing an average IoU gain of 0.14 ($\sim 20\%$ relative). These results show that combining learned proposals with targeted program-space search is essential for high-fidelity CAD reconstruction.

Surface accuracy. Chamfer Distance follows the same trend. StepCAD achieves the lowest error on all datasets (e.g., 0.0263 vs. 0.0283 on ABC Hard, and 0.0420 vs. 0.0446 on Organic), indicating improvements not only in volumetric overlap but also in fine geometric detail.

Disentangling training data, policy, and optimization. Table 3 adds a single-shot Qwen2.5 model fine-tuned on ARCADE-1.5M using a CAD-Recode-style point-cloud-to-program setup. It reaches 0.644 IoU and 0.041 CD without intermediate-geometry conditioning or search, compared with 0.568 IoU for the released CADEvolve checkpoint. This demonstrates that the corpus supports a conventional generator, although differences in model and training setup prevent attributing that cross-model gap solely to data. Within StepCAD, removing previous-geometry tokens reduces IoU from 0.682 to 0.632. The full policy exceeds the single-shot baseline by 0.038 IoU, and program-space search further improves the policy from 0.682 to 0.816 (+0.134). Together, the experiments distinguish the benefits of an operation-rich corpus, intermediate-state conditioning, and explicit refinement without treating the released-checkpoint comparison as a controlled data-only ablation.

Table 3: **Training data, state conditioning, and search.** All variants except the released CADEvolve checkpoint use ARCADE-1.5M. The single-shot baseline predicts a complete program without intermediate geometry.

Variant	CD $\downarrow$	IoU $\uparrow$	VSR (%) $\uparrow$
CADEvolve (public checkpoint)	0.056	0.568	97
Single-shot Qwen2.5 on ARCADE-1.5M	0.041	0.644	100
Π -StepCAD w/o previous-geometry tokens	0.045	0.632	100
Π -StepCAD	0.028	0.682	100
StepCAD	0.025	0.816	100

Additional evaluation: fidelity, search transfer, and compactness. Appendix K complements the main benchmark results with high-IoU success rates, entity-level geometric accuracy, search transferability, and program compactness. StepCAD achieves $\text{IoU} \geq 0.95$ on 39.9% of CADBench samples and $\text{IoU} \geq 0.99$ on 15.3%, compared with 27.2% and 7.4% for the strongest baseline, CADReasoner (Table 9). It also improves vertex, edge, and face accuracy from 15.7, 9.7, and 6.4 for the strongest baseline, CADEvolve, to 20.8, 12.6, and 8.1, respectively (Table 11). Applying

our search to CAD-Recode programs on 1,200 samples improves mean IoU from 0.597 to 0.745 without retraining (Table 10). Refinement increases mean program size from 16.2 to 31.7 counted operations as reconstruction fidelity improves (Table 12); on DeepCAD, the refined programs have an overall recovered/reference length ratio of 0.91 at 0.89 IoU (Table 13). These evaluations distinguish geometric fidelity and compactness from recovery of a unique parameterization or design intent.

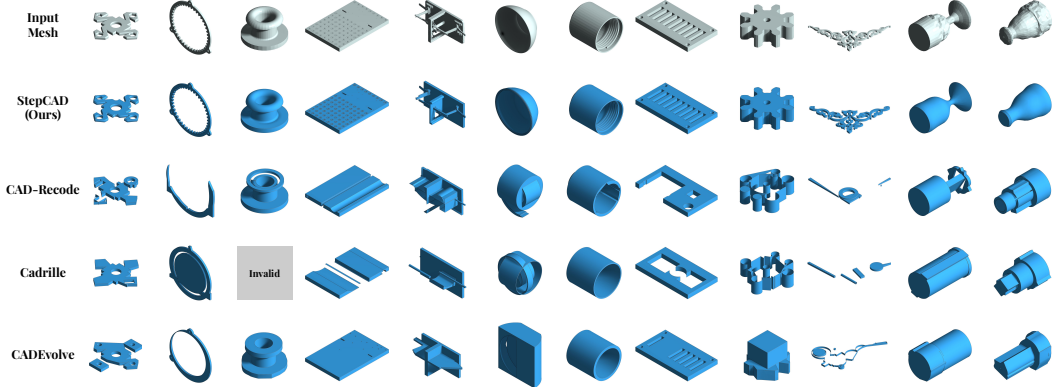


Figure 3: **Qualitative mesh-to-CAD reconstruction results.** Rows show input mesh, StepCAD, CAD-Recode, Cadrille, and CADEvolve; ‘Invalid’ denotes a failed/non-executable reconstruction.

Qualitative results. Appendix E provides construction trajectories (Fig. 8), competing search edits and finishing-feature examples (Fig. 10), failure cases (Fig. 11), evaluation inputs (Fig. 7), and expanded policy-versus-refinement comparisons (Fig. 9). Figure 3 shows reconstructions across datasets. StepCAD produces CAD programs that closely match target geometry while preserving both global structure and fine details.

Impact of ARCADE-1.5M and operation diversity. A key factor behind these gains is the broader operation coverage of **ARCADE-1.5M**, including fillets, chamfers, lofts, and revolves. Earlier sketch-extrude corpora offer narrower coverage, while recent corpora such as CADEvolve and Zero-to-CAD also contain finishing and other solid operations. Their presence in training data should be distinguished from their frequency in generated programs. As a result, StepCAD generates more diverse operations. On the **Organic** subset, lofts appear in 30.4% of generated programs versus 3.0% for CADEvolve, while revolves appear in 23.2% versus 0.8%. Prior methods rely overwhelmingly on extrusions. (Appendix D)

This diversity is critical for reconstructing geometries that cannot be represented well with sketch-extrude primitives alone. For example, in the last column of Fig. 3, StepCAD uses loft operations to recover smooth non-planar transitions, achieving higher fidelity (**IoU**: 0.83, **CD**: 0.066) than CADEvolve (**IoU**: 0.66, **CD**: 0.108) and Cadrille (**IoU**: 0.67, **CD**: 0.101), which primarily rely on extrusion-based constructions. These results demonstrate that operation diversity is essential for faithful reconstruction of complex geometry.

Applications and one-to-many CAD generation. Beyond reconstruction, StepCAD enables several downstream applications enabled by its state-conditioned generative formulation. As shown in Appendix B and Fig. 4, II-StepCAD can generate diverse editable CAD variants from a single reference mesh while preserving the overall structure, enabling rapid design exploration and parametric shape variation. StepCAD can also be integrated into image-to-CAD pipelines (Appendix B, Fig. 5) by reconstructing parametric CAD programs from meshes generated by image-to-3D models such as Hunyuan3D [6]. Finally, Appendix C and Fig. 6 highlight the inherently one-to-many nature of mesh-to-CAD reconstruction: the policy can generate multiple valid CAD programs with different operation sequences that produce nearly identical geometry. This suggests that StepCAD learns a distribution over plausible CAD construction strategies rather than memorizing a single deterministic reconstruction process.

Overall, these results demonstrate the effectiveness of combining state-conditioned generative policies with geometry-aware optimization for robust and accurate CAD program synthesis.

Additional ablations. Additional ablations are provided in the appendix, including the effects of Qwen model scale and point-cloud encoder choice (Appendix A.2, Table 5); policy inputs, Point-MAE fine-tuning, and point sampling strategy (Appendix A.1, Table 4); and the contribution of each local edit within the tree-search optimization procedure (Appendix A.3, Table 6).

5 Limitations and Future Work

Computation and search. Refinement takes approximately one minute per sample and relies on a fixed set of local edits. Its bounded parameter neighborhoods and finite beam may miss a useful correction or fail to recover from a poor seed. The 12-action policy cap limits generation even though one action may contain multiple counted operations; training sequences longer than this cap do not establish equivalent inference-time capacity. **Geometric versus semantic fidelity.** IoU and CD measure shape agreement, not recovery of original dimensions, design intent, constraints, or feature semantics. In particular, removing a misparameterized but meaningful fillet may improve IoU if a better radius is not found. **Selection and editability.** Coarse edge and directional face selectors cannot address arbitrary individual entities. Programs are executable and their code parameters can be edited, but we do not establish constraint preservation or user editing success. Program compactness is only one aspect of editability. **Data and generalization.** Synthetic training introduces a domain gap, and alternative valid histories complicate comparison with a reference B-rep. Future work should combine aligned point-cloud/B-rep representations, entity-specific operators, feature-aware objectives, constraint modeling, and user studies, while improving search efficiency. **Societal impact.** Reconstruction can facilitate replication of proprietary designs and change designer workflows, motivating attribution and human oversight.

6 Conclusion

We presented **StepCAD**, a generative optimization approach for mesh-to-CAD reconstruction that combines a state-conditioned CAD policy with geometry-aware program refinement. By reformulating CAD generation as optimization in program space, our approach overcomes the limitations of one-shot sequence prediction and achieves SOTA performance across multiple benchmarks. We further introduced **ARCADE-1.5M**, a large-scale dataset of CAD programs with diverse operations and long-horizon construction sequences, enabling effective training of state-conditioned models. Our results demonstrate that combining learned generation with explicit geometric optimization leads to substantial improvements in fidelity, robustness, and generalization. More broadly, this work highlights the importance of integrating generative models with search and optimization for structured program synthesis tasks. We believe this paradigm can be extended beyond CAD reconstruction to applications such as image-to-CAD generation, interactive design, and automated engineering workflows.

References

- [1] Ahmet Serdar Karadeniz, Dimitrios Mallis, Danila Rukhovich, Kseniya Cherenkova, Anis Kacem, and Djamila Aouada. MiCADangelo: Fine-grained reconstruction of constrained CAD models from 3d scans. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [2] Danila Rukhovich, Elona Dupont, Dimitrios Mallis, Kseniya Cherenkova, Anis Kacem, and Djamila Aouada. Cad-recode: Reverse engineering cad code from point clouds. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9801–9811, 2025.
- [3] Hongyuan Ren, Bo Xia, Wenrui Wang, Xueqian Chen, Yang Liu, Pingzhang Zhou, and Jianbin Du. Amrto: Automated cad model reconstruction of topology optimization result. *Computer Methods in Applied Mechanics and Engineering*, 435:117673, 2025.
- [4] Abdenmour Amroune, Jean-Christophe Cuilliere, and Vincent Francois. Automated lofting-based reconstruction of cad models from 3d topology optimization results. *Computer-Aided Design*, 145:103183, 2022.
- [5] Jianfeng Xiang, Zelong Lv, Sicheng Xu, Yu Deng, Ruicheng Wang, Bowen Zhang, Dong Chen, Xin Tong, and Jiaolong Yang. Structured 3d latents for scalable and versatile 3d generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 21469–21480, 2025.
- [6] Zibo Zhao, Zeqiang Lai, Qingxiang Lin, Yunfei Zhao, Haolin Liu, Shuhui Yang, Yifei Feng, Mingxin Yang, Sheng Zhang, Xianghui Yang, et al. Hunyuan3d 2.0: Scaling diffusion models for high resolution textured 3d assets generation. *arXiv preprint arXiv:2501.12202*, 2025.
- [7] Yuan Li, Cheng Lin, Yuan Liu, Xiaoxiao Long, Chenxu Zhang, Ningna Wang, Xin Li, Wenping Wang, and Xiaohu Guo. Caddreamer: Cad object generation from single-view images. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 21448–21457, 2025.
- [8] Rundi Wu, Chang Xiao, and Changxi Zheng. Deepcad: A deep generative network for computer-aided design models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- [9] Xiang Xu, Pradeep Kumar Jayaraman, Joseph G Lambourne, Karl DD Willis, and Yasutaka Furukawa. Hierarchical neural coding for controllable cad model generation. *arXiv preprint arXiv:2307.00149*, 2023.
- [10] Maksim Kolodiaznyi, Denis Tarasov, Dmitrii Zhemchuzhnikov, Alexander Nikulin, Ilya Zisman, Anna Vorontsova, Anton Konushin, Vladislav Kurenkov, and Danila Rukhovich. cadrille: Multi-modal cad reconstruction with online reinforcement learning. *arXiv preprint arXiv:2505.22914*, 2025.
- [11] Jingwei Xu, Chenyu Wang, Zibo Zhao, Wen Liu, Yi Ma, and Shenghua Gao. Cad-mllm: Unifying multimodality-conditioned cad generation with mllm. *arXiv preprint arXiv:2411.04954*, 2024.
- [12] Mohammadmehdi Ataei, Farzaneh Askari, Kamal Rahimi Malekshan, and Pradeep Kumar Jayaraman. Zero-to-cad: Agentic synthesis of interpretable cad programs at million-scale without real data. *arXiv preprint arXiv:2604.24479*, 2026.
- [13] Maksim Elistratov, Marina Barannikov, Gregory Ivanov, Valentin Khrulkov, Anton Konushin, Andrey Kuznetsov, and Dmitrii Zhemchuzhnikov. Cadevolve: Creating realistic cad via program evolution. *arXiv preprint arXiv:2602.16317*, 2026.
- [14] Matej Balog, Alexander L Gaunt, Marc Brockschmidt, Sebastian Nowozin, and Daniel Tarlow. Deepcoder: Learning to write programs. *arXiv preprint arXiv:1611.01989*, 2016.
- [15] Kevin Ellis, Maxwell Nye, Yewen Pu, Felix Sosa, Josh Tenenbaum, and Armando Solar-Lezama. Write, execute, assess: Program synthesis with a repl. *Advances in Neural Information Processing Systems*, 32, 2019.

- [16] Xinyun Chen, Chang Liu, and Dawn Song. Execution-guided neural program synthesis. In *International Conference on Learning Representations*, 2018.
- [17] Lyle Regenwetter, Cyril Picard, Amin Heyrani Nobari, Akash Srivastava, and Faez Ahmed. Generative optimization: A perspective on ai-enhanced problem solving in engineering. *arXiv preprint arXiv:2412.13281*, 2024.
- [18] Anna C Doris, Jacob Thomas Sony, Ghadi Nehme, Era Syla, Amin Heyrani Nobari, and Faez Ahmed. Cadbench: A multimodal benchmark for ai-assisted cad program generation. *arXiv preprint arXiv:2605.10873*, 2026.
- [19] Sebastian Koch, Albert Matveev, Zhongshi Jiang, Francis Williams, Alexey Artemov, Evgeny Burnaev, Marc Alexa, Denis Zorin, and Daniele Panozzo. Abc: A big cad model dataset for geometric deep learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9601–9611, 2019.
- [20] Karl DD Willis, Yewen Pu, Jieliang Luo, Hang Chu, Tao Du, Joseph G Lambourne, Armando Solar-Lezama, and Wojciech Matusik. Fusion 360 gallery: A dataset and environment for programmatic cad construction from human design sequences. *ACM Transactions on Graphics (TOG)*, 40(4):1–24, 2021.
- [21] Sangpil Kim, Hyung-gun Chi, Xiao Hu, Qixing Huang, and Karthik Ramani. A large-scale annotated mechanical components benchmark for classification and retrieval tasks with deep neural networks. In *European conference on computer vision*, pages 175–191. Springer, 2020.
- [22] Matt Deitke, Dustin Schwenk, Jordi Salvador, Luca Weihs, Oscar Michel, Eli VanderBilt, Ludwig Schmidt, Kiana Ehsani, Aniruddha Kembhavi, and Ali Farhadi. Objaverse: A universe of annotated 3d objects. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 13142–13153, 2023.
- [23] Changqi He, Shuhan Zhang, Liguozhang, and Jiajun Miao. Cad-coder: Text-guided cad files code generation. *arXiv preprint arXiv:2505.08686*, 2025.
- [24] Ruiyu Wang, Yu Yuan, Shizhao Sun, and Jiang Bian. Text-to-cad generation through infusing visual feedback in large language models. *arXiv preprint arXiv:2501.19054*, 2025.
- [25] Mohammad S Khan, Sankalp Sinha, Talha U Sheikh, Didier Stricker, Sk A Ali, and Muhammad Z Afzal. Text2cad: Generating sequential cad designs from beginner-to-expert level text prompts. *Advances in Neural Information Processing Systems*, 37:7552–7579, 2024.
- [26] Yujia Liu, Anton Obukhov, Jan Dirk Wegner, and Konrad Schindler. Point2cad: Reverse engineering cad models from 3d point clouds. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3763–3772, 2024.
- [27] Shane Larsen and C Greg Jensen. Converting topology optimization results into parametric cad models. *Computer-Aided Design and Applications*, 6(3):407–418, 2009.
- [28] Nomi Yu, Md Ferdous Alam, A John Hart, and Faez Ahmed. Gencad-3d: Cad program generation using multimodal latent space alignment and synthetic dataset balancing. In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, volume 89220, page V03AT03A015. American Society of Mechanical Engineers, 2025.
- [29] Xiang Xu, Karl DD Willis, Joseph G Lambourne, Chin-Yi Cheng, Pradeep Kumar Jayaraman, and Yasutaka Furukawa. Skexgen: Autoregressive generation of cad construction sequences with disentangled codebooks. *arXiv preprint arXiv:2207.04632*, 2022.
- [30] King Yiu Brandon Man, Ghadi Nehme, Md Ferdous Alam, and Faez Ahmed. Videocad: A dataset and model for learning long-horizon 3d cad ui interactions from video. *Advances in Neural Information Processing Systems*, 38, 2026.
- [31] Cheng Chen, Jiacheng Wei, Tianrun Chen, Chi Zhang, Xiaofeng Yang, Shangzhan Zhang, Bingchen Yang, Chuan-Sheng Foo, Guosheng Lin, Qixing Huang, et al. Cadrafter: Generating computer-aided design models from unconstrained images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11073–11082, 2025.

- [32] Anna C Doris, Md Ferdous Alam, Amin Heyrani Nobari, and Faez Ahmed. Cad-coder: An open-source vision-language model for computer-aided design code generation. In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, volume 89220, page V03AT03A031. American Society of Mechanical Engineers, 2025.
- [33] Soslan Kabisov, Vsevolod Kirichuk, Andrey Volkov, Gennadii Savrasov, Marina Barannikov, Anton Konushin, Andrey Kuznetsov, and Dmitrii Zhemchuzhnikov. Cadreasoner: Iterative program editing for cad reverse engineering. *arXiv preprint arXiv:2603.29847*, 2026.
- [34] Ansong Ni, Srini Iyer, Dragomir Radev, Veselin Stoyanov, Wen-tau Yih, Sida Wang, and Xi Victoria Lin. Lever: Learning to verify language-to-code generation with execution. In *International Conference on Machine Learning*, pages 26106–26128. PMLR, 2023.
- [35] Xinyun Chen, Maxwell Lin, Nathanael Schärli, and Denny Zhou. Teaching large language models to self-debug. *arXiv preprint arXiv:2304.05128*, 2023.
- [36] Shun Zhang, Zhenfang Chen, Yikang Shen, Mingyu Ding, Joshua B Tenenbaum, and Chuang Gan. Planning with large language models for code generation. *arXiv preprint arXiv:2303.05510*, 2023.
- [37] Dacheng Qi, Chenyu Wang, Jingwei Xu, Tianzhe Chu, Zibo Zhao, Wen Liu, Wenrui Ding, Yi Ma, and Shenghua Gao. Pointer-cad: Unifying b-rep and command sequences via pointer-based edges & faces selection. *arXiv preprint arXiv:2603.04337*, 2026.
- [38] Jiahao Li, Qingwang Zhang, Qiuyu Chen, Guozhan Qiu, Yunzhong Lou, and Xiangdong Zhou. Towards high-fidelity cad generation via llm-driven program generation and text-based b-rep primitive grounding. *arXiv preprint arXiv:2603.11831*, 2026.
- [39] Xueyang Li, Jiahao Li, Yu Song, Yunzhong Lou, and Xiangdong Zhou. Seek-cad: A self-refined generative modeling for 3d parametric cad using local inference via deepseek. In *International Conference on Learning Representations*, volume 2026, pages 94278–94298, 2026.
- [40] Dacheng Qi, Chenyu Wang, Jingwei Xu, Yi Ma, and Shenghua Gao. Pointer-cad v2: Plan-then-construct cad generation with dimension-aware parametric precision. *arXiv preprint arXiv:2606.29301*, 2026.
- [41] Ghadi Nehme, Eamon Whalen, and Faez Ahmed. Cadfit: Precise mesh-to-cad program generation with hybrid optimization. *arXiv preprint arXiv:2605.01171*, 2026.
- [42] Xiang Xu, Joseph Lambourne, Pradeep Jayaraman, Zhengqing Wang, Karl Willis, and Yasutaka Furukawa. Brep-gen: A b-rep generative diffusion model with structured latent geometry. *ACM Transactions on Graphics (TOG)*, 43(4):1–14, 2024.
- [43] Mingi Lee, Dongsu Zhang, Clément Jambon, and Young Min Kim. Brepdiff: Single-stage b-rep diffusion model. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, pages 1–11, 2025.
- [44] Mikaela Angelina Uy, Yen-Yu Chang, Minhyuk Sung, Purvi Goel, Joseph G Lambourne, Tolga Birdal, and Leonidas J Guibas. Point2cyl: Reverse engineering 3d objects from point clouds to extrusion cylinders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11850–11860, 2022.
- [45] Tao Du, Jeevana Priya Inala, Yewen Pu, Andrew Spielberg, Adriana Schulz, Daniela Rus, Armando Solar-Lezama, and Wojciech Matusik. Inversecsg: Automatic conversion of 3d models to csg trees. *ACM Transactions on Graphics (TOG)*, 37(6):1–16, 2018.
- [46] CadQuery contributors. CadQuery. <https://doi.org/10.5281/zenodo.14590990>, October 2025. <https://doi.org/10.5281/zenodo.14590990>.
- [47] Tianrun Chen, Chunan Yu, Yuanqi Hu, Jing Li, Tao Xu, Runlong Cao, Lanyun Zhu, Ying Zang, Yong Zhang, Zejian Li, et al. Img2cad: Conditioned 3-d cad model generation from single image with structured visual geometry. *IEEE Transactions on Industrial Informatics*, 2025.

- [48] Yatian Pang, Eng Hock Francis Tay, Li Yuan, and Zhenghua Chen. Masked autoencoders for 3d point cloud self-supervised learning. *World Scientific Annual Review of Artificial Intelligence*, 1:2440001, 2023.
- [49] Binyuan Hui, Jian Yang, Zeyu Cui, Jiayi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Keming Lu, et al. Qwen2. 5-coder technical report. *arXiv preprint arXiv:2409.12186*, 2024.
- [50] Manuel Contero, David Pérez-López, Pedro Company, and Jorge D Camba. A quantitative analysis of parametric cad model complexity and its relationship to perceived modeling complexity. *Advanced Engineering Informatics*, 56:101970, 2023.
- [51] Yujia Zhang, Xiaoyang Wu, Yunhan Yang, Xianzhe Fan, Han Li, Yuechen Zhang, Zehao Huang, Naiyan Wang, and Hengshuang Zhao. Utonia: Toward one encoder for all point clouds. *arXiv preprint arXiv:2603.03283*, 2026.

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A Ablations

A.1 Which Components of Π -StepCAD Matter Most?

Table 4: **Ablation study of StepCAD.** CD $\downarrow$ and IoU $\uparrow$ evaluate reconstruction quality; VSR denotes valid shape rate.

Variant	CD $\downarrow$	IoU $\uparrow$	VSR (%) $\uparrow$
Π -StepCAD w/o prompt conditioning	–	–	0
Π -StepCAD w/o FPS point sampling	0.209	0.105	100
Π -StepCAD w/o input geometry tokens	0.144	0.209	100
Π -StepCAD w/o Point-MAE fine-tuning	0.093	0.316	100
CADEvolve (public checkpoint)	0.056	0.568	97
Single-shot Qwen2.5 on ARCADE-1.5M	0.041	0.644	100
Π -StepCAD w/o previous-geometry tokens	0.045	0.632	100
Π-StepCAD	0.028	0.682	100
StepCAD	0.025	0.816	100

A.2 How Do Model Scale and Geometry Encoders Affect Performance?

Table 5: **Ablation study for model size and point cloud encoder for Π -StepCAD.** Best validation loss achieved within the first 10K training steps. Lower is better.

Variant	Best Val Loss at 10K steps $\downarrow$
Π -StepCAD (0.5B + Point-MAE)	0.3559
Π -StepCAD (1.5B + Point-MAE)	<u>0.3433</u>
Π-StepCAD (3B + Point-MAE)	0.3419
Π -StepCAD (1.5B + Utonia [51])	0.3450

Although the 3B model achieves the lowest early validation loss, its gain over the 1.5B model is marginal (0.0014), so we use the 1.5B Point-MAE configuration as the default due to its favorable compute-performance trade-off.

A.3 Which Local Edits Contribute Most to Geometry-Aware Search?

Table 6: **Local edit-level ablation of the tree search.** Mean metrics;

Win-rate: fraction of beam steps where this phase’s output is the rank-0 candidate (i.e. the new beam[0]).

Realized Δ IoU: average per-sample IoU lift attributed to this phase via winner-attribution at each step; the four phase rows sum to the *Full search* row by construction.

% share: the phase’s Realized Δ divided by the Full Realized Δ .

Cost: mean wall-time per sample.

Local Edit	Win-rate	Realized Δ	% share	Cost (s)
Sketch Modification	26.0%	+0.059	41.5%	12.5
Complement Cut	24.4%	+0.043	30.3%	17.0
Continuous refinement	36.6%	+0.023	16.2%	28.4
Operation skipping	13.0%	+0.017	12.0%	8.1
Full search	100%	+0.142	100%	66.0

B StepCAD: Applications

B.1 Variants Generation

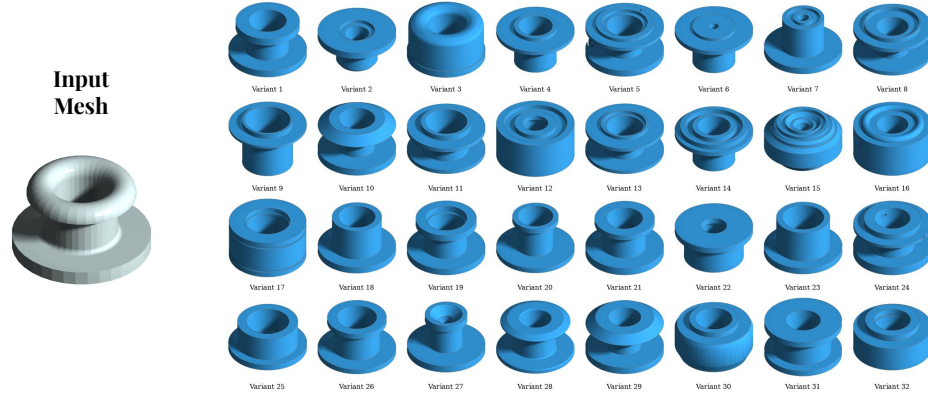


Figure 4: **Design variation generation with II-StepCAD.** Given an input mesh, II-StepCAD can generate diverse parametric CAD variants that preserve the overall structure while modifying geometric details and construction choices. This demonstrates a potential application of the learned policy for CAD design exploration and rapid generation of editable shape variations from a single reference geometry.

B.2 Image-to-CAD

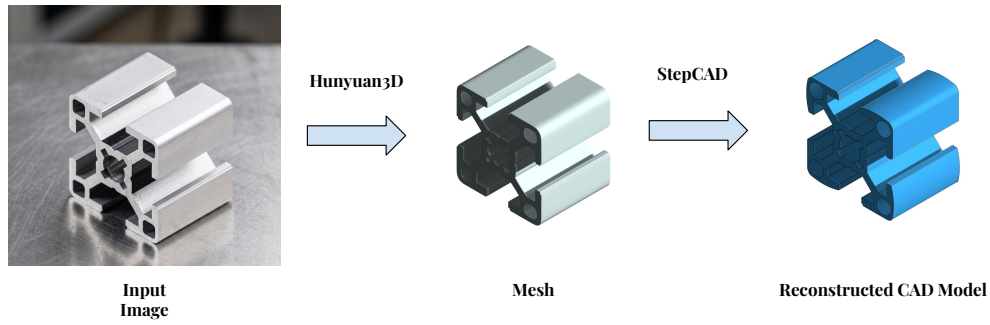


Figure 5: **Image-to-CAD reconstruction using StepCAD.** An input image is first converted into a 3D mesh using Hunyuan3D [6], after which StepCAD reconstructs an editable parametric CAD program from the generated geometry. This demonstrates how StepCAD can be integrated into image-to-CAD pipelines to transform unstructured visual inputs into structured CAD representations.

C One-to-Many Mapping: From Geometry to CAD Construction Sequence

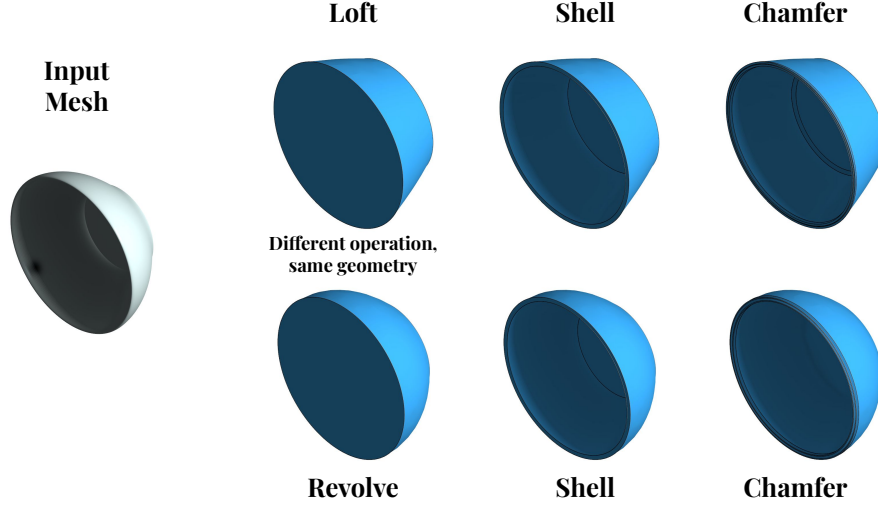


Figure 6: **One-to-many CAD program generation from a single input mesh.** Given the same target geometry, StepCAD generates multiple valid CAD programs with different operation sequences. In this example, the model reconstructs the shape using either a loft-based or revolve-based strategy, followed by shell and chamfer operations, producing nearly identical final geometry. This demonstrates that the policy learns a distribution over valid CAD construction procedures rather than a single deterministic reconstruction.

D Analysis of Generated CAD Operation Distributions

Table 7: **Operation occurrence across generated programs (%)**. Values indicate the percentage of generated CAD programs that contain at least one instance of the given operation.

Model	Extrude	Revolve	Sweep	Loft	Fillet	Chamfer	Shell	Mirror
StepCAD	89.35	16.86	7.75	11.48	9.60	6.24	5.11	2.82
CADEvolve	81.10	0.20	0.00	1.20	3.30	1.80	0.20	0.00
Cadrille	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CAD-Recode	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 8: **Operation occurrence across generated programs (%)**. Values indicate the percentage of generated CAD programs that contain at least one instance of the given operation.

Benchmark	Model	Extrude	Revolve	Sweep	Loft	Fillet	Chamfer	Shell	Mirror
DeepCAD	StepCAD	94.5	11.3	4.4	4.9	4.1	1.7	2.9	2.2
	CADEvolve	80.9	0.0	0.0	0.0	0.1	0.0	0.4	0.0
Fusion360	StepCAD	86.7	19.9	10.2	12.5	10.0	4.9	3.3	3.1
	CADEvolve	81.7	0.0	0.0	0.8	3.1	1.0	0.1	0.0
ABC	StepCAD	92.0	14.4	6.1	8.4	9.9	5.5	7.4	2.7
	CADEvolve	82.9	0.0	0.0	0.6	4.8	1.3	0.2	0.0
Mechanical	StepCAD	75.7	33.0	6.1	32.2	23.3	17.6	5.3	2.3
	CADEvolve	70.2	0.0	0.0	2.5	5.0	6.8	0.0	0.0
Organic	StepCAD	79.6	23.2	21.8	30.4	12.9	20.4	6.5	4.8
	CADEvolve	88.1	0.8	0.1	3.0	1.9	0.6	0.2	0.0

E Construction Trajectories, Search Behavior, and Qualitative Evaluation

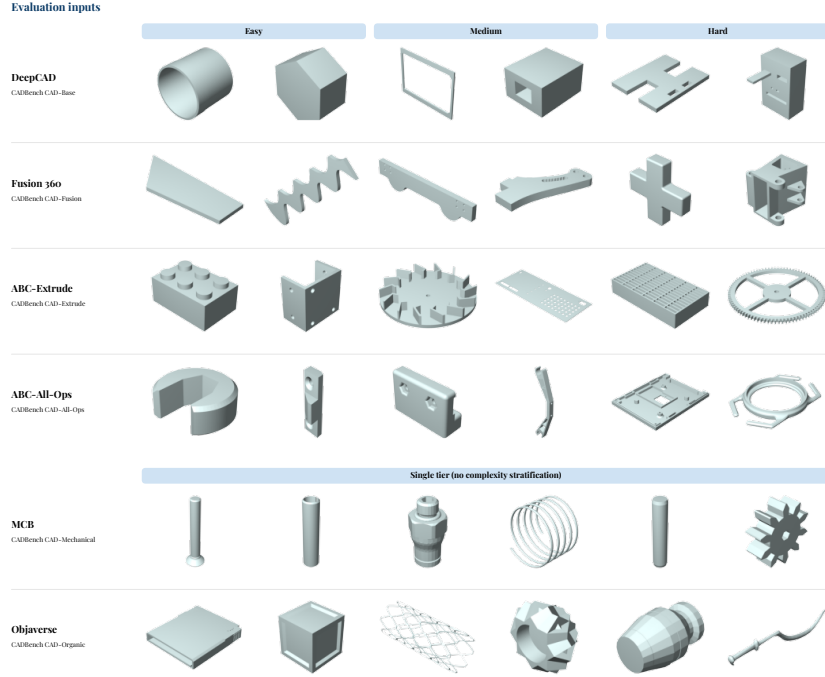


Figure 7: **Evaluation-input gallery.** Example meshes from DeepCAD, Fusion 360, ABC-Extrude, ABC-All-Ops, MCB, and Objaverse, using the category labels supplied with the gallery.

Construction trajectories: policy actions, then tree-search refinement

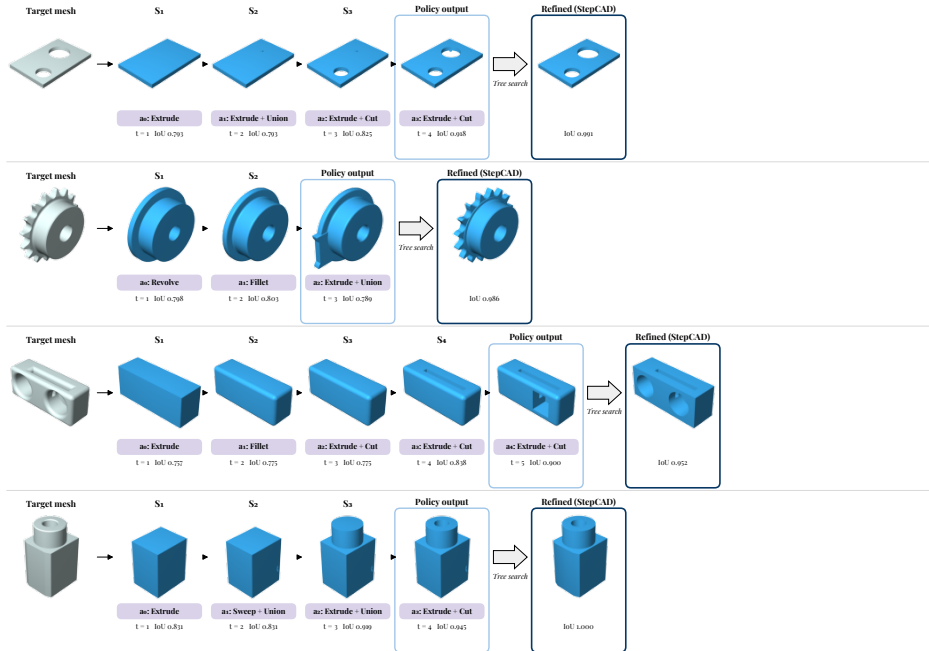


Figure 8: **Construction trajectories followed by refinement.** Four examples show the target mesh, successive intermediate solids, the policy output, and the refined StepCAD reconstruction.

	DeepCAD Easy	DeepCAD Medium	DeepCAD Hard	Fusion 360 Easy	Fusion 360 Medium	Fusion 360 Hard	ABC- Extrude Easy
Input Mesh							
Policy Only							
	<i>IoU</i>	0.03	0.39	0.90	0.92	0.48	0.99
StepCAD (Ours)							
	<i>IoU</i>	0.89	0.72	0.94	0.93	0.98	1.00

	ABC- Extrude Medium	ABC- Extrude Hard	ABC- All-Ops Easy	ABC- All-Ops Medium	ABC- All-Ops Hard	MCB	Objaverse
Input Mesh							
Policy Only							
	<i>IoU</i>	0.46	0.83	0.83	0.92	0.45	0.87
StepCAD (Ours)							
	<i>IoU</i>	0.82	0.95	0.97	0.95	0.74	0.95

	DeepCAD Easy	DeepCAD Medium	DeepCAD Hard	Fusion 360 Easy	Fusion 360 Medium	Fusion 360 Hard	ABC- Extrude Easy
Input Mesh							
Policy Only							
	<i>IoU</i>	0.97	0.84	0.89	0.92	0.86	0.37
StepCAD (Ours)							
	<i>IoU</i>	0.99	0.91	0.98	0.93	0.99	0.92

	ABC- Extrude Medium	ABC- Extrude Hard	ABC- All-Ops Easy	ABC- All-Ops Medium	ABC- All-Ops Hard	MCB	Objaverse
Input Mesh							
Policy Only							
	<i>IoU</i>	0.91	0.73	0.79	0.84	0.91	0.71
StepCAD (Ours)							
	<i>IoU</i>	0.96	0.97	0.89	0.87	0.99	0.89

Figure 9: Expanded qualitative reconstructions

Search edits and feature preservation



Figure 10: **Competing search edits and finishing-feature examples.** The upper panels compare four edit proposals from a common parent, with IoU and retained/rejected status. The lower-left panels compare parameter correction with removal of a chamfer or fillet; correction yields higher IoU in these examples. The lower-right panel shows a beam-search trace with retained and rejected candidates.

Failure cases

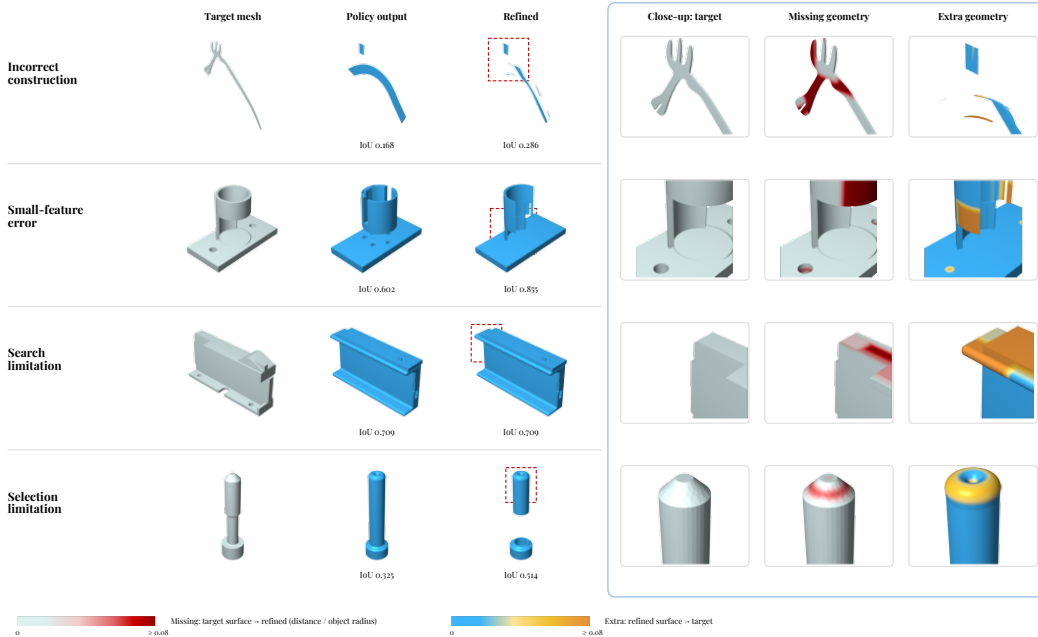


Figure 11: **Residual errors and failure cases.** Target meshes, policy outputs, and refined reconstructions are shown with IoU.

F Metrics

We report the following metrics:

- **Intersection-over-Union (IoU):** volumetric overlap between predicted and target shapes.

$$\text{IoU} = \frac{V(\hat{S} \cap S)}{V(\hat{S} \cup S)}$$

- **Chamfer Distance (CD):** surface distance between point clouds (lower is better).

$$\text{CD}(\hat{P}, P) = \frac{1}{2} \left(\frac{1}{|\hat{P}|} \sum_{x \in \hat{P}} \min_{y \in P} \|x - y\|_2^2 + \frac{1}{|P|} \sum_{y \in P} \min_{x \in \hat{P}} \|y - x\|_2^2 \right)$$

- **Valid Shape Rate (VSR):** percentage of generated programs that execute successfully and produce valid geometry.

G Dataset Statistics

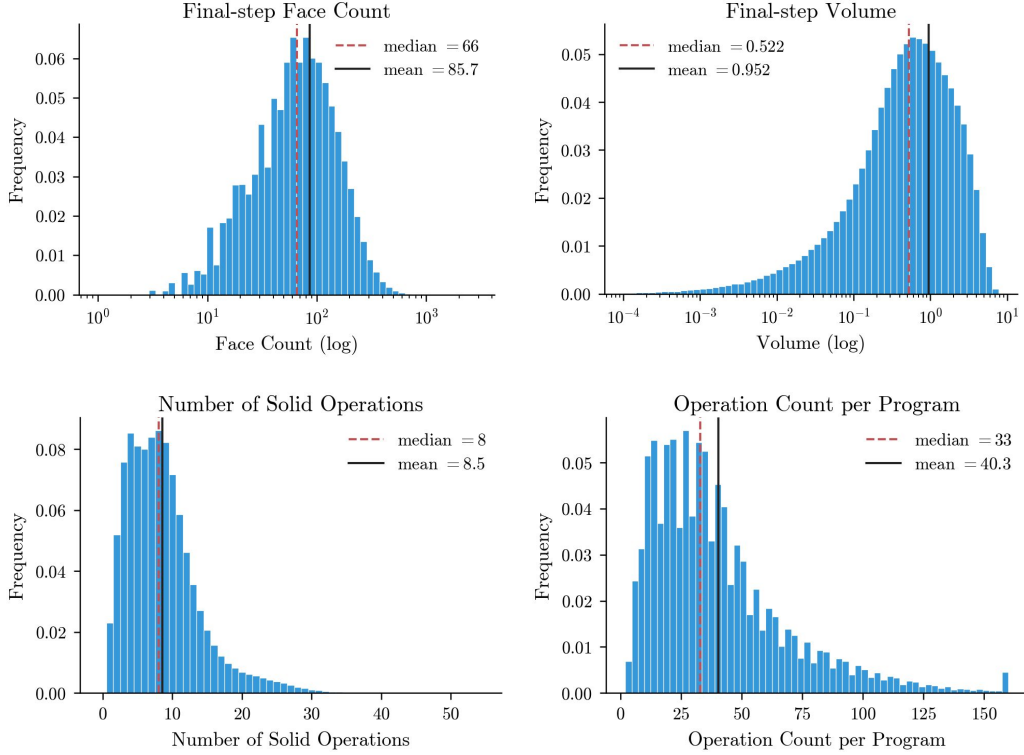


Figure 12: **ARCADE-1.5M dataset statistics.** We report distributions of face count, volume, solid operations and total operations per program. The dataset spans a wide range of shapes and scales, while exhibiting long-horizon construction sequences and non-trivial solid modeling depth. Operation counts range from 3 to 150+, with a median of 33 and a mean of 40.3. Dashed and solid lines denote median and mean, respectively.

H Dataset Generation Details

H.1 Sketch Database

The fundamental source of geometric diversity is a curated database of 2-D sketch loop primitives derived from real engineering CAD models. Each sketch entry is a dictionary containing a list of *primitives*, where every primitive is one of four types:

- **Line:** a pair of (x, y) endpoints.
- **Arc:** three control points (start, mid, end) defining a circular arc, encoded as a `threePointArc`.
- **Circle:** a centre point and a radius.
- **Spline:** an ordered list of (x, y) control points defining a B-spline curve.

At runtime, `CADFeatureTreeGenerator.get_random_sketch()` draws a uniformly random entry from this index and loads it on demand, with an in-memory LRU cache to avoid repeated disk I/O.

H.2 Sketch Placement and Scaling

Before a sketch is used to build a solid it must be fitted into a caller-specified 3-D axis-aligned bounding box (*bbox*). The procedure is:

1. **Compute sketch 2-D bbox.** All primitive control points are collected and the tight axis-aligned bounding box $[s_x^{\min}, s_y^{\min}, s_x^{\max}, s_y^{\max}]$ is computed.
2. **Select a face of the target 3-D bbox.** The six bbox faces are enumerated with their outward normals and local x -directions. One face is chosen uniformly at random as the workplane for the sketch.
3. **Compute uniform scale and 2-D translation.** The sketch is uniformly scaled so that it fills a random fraction f of the face's 2-D extent, then translated so its centre coincides with the face centre:
$$s = f \cdot \min\left(\frac{2T_w}{S_w}, \frac{2T_h}{S_h}\right), \quad c_x = \frac{T_x^{\min} + T_x^{\max}}{2}, \quad c_y = \frac{T_y^{\min} + T_y^{\max}}{2}.$$
4. **Apply transform to primitives.** Every control point (p_x, p_y) is mapped to $(p_x \cdot s + c_x, p_y \cdot s + c_y)$ before being written into the generated code.

H.3 Primitive Code Generation

Given a placed-and-scaled sketch and a target 3-D bbox, `generate_cadquery_code()` emits CadQuery Python code for one of four volumetric operations.

Extrude. A named workplane is created from `cq.Plane(origin=..., normal=..., xDir=...)`, aligned with the chosen bbox face. The sketch primitives are emitted in order: lines as `.moveTo(x0,y0).lineTo(x1,y1)` (first) or `.lineTo(x1,y1)` (subsequent); arcs as `.threePointArc((x1,y1),(x2,y2))`; circles as `.moveTo(cx,cy).circle(r)`; splines as `.spline([(x0,y0),...])`. A `.close()` seals the contour and `.extrude(d)` creates the solid, where d fills 85–100% of the bbox dimension normal to the face. Example output:

```
plane_0 = cq.Plane(origin=(0,0,0), normal=(0,0,1), xDir=(1,0,0))
sketch_0 = cq.Workplane(plane_0)
sketch_0 = sketch_0.moveTo(-1.2,-0.8).lineTo(1.2,-0.8)
sketch_0 = sketch_0.threePointArc((1.5,0.0),(1.2,0.8))
sketch_0 = sketch_0.lineTo(-1.2,0.8).close()
solid_0 = sketch_0.extrude(2.5)
```

Revolve. Workplane and sketch emission are identical to extrude. After `.close()`, the solid is created with `.revolve(angleDegrees= $\theta$ , axisStart=(ax,ay), axisEnd=(bx,by))`, where $\theta \in \{90, 180, 270, 360\}$ is chosen uniformly and the axis is either a line extracted from the sketch primitives (50% probability) or an edge of the face bbox. With 25% probability a circle replaces the database sketch profile.

Sweep (spline path). A 2-D spline path is generated by sampling one of ten named archetypes — *open*: arc, S-curve, U-curve, L-curve, wave, diagonal; *closed*: ellipse, rounded rectangle, teardrop, kidney — and emitted as:

```
path_0 = cq.Workplane('XY').spline([(x0,y0),(x1,y1),...])
```

(with `.close()` for closed loops). The profile cross-section on the start face is emitted identically to the extrude case, then swept with `profile_0.sweep(path_0)`.

Loft (offset). A single workplane is created. The sketch is drawn $N \in [3, 7]$ times at successive `.workplane(offset= $\delta_i$ )` steps, each with a per-profile scale factor that creates a tapered or gradually varying cross-section. The chain is closed with `.loft()`.

Loft (transformed). The base profile is drawn on the start workplane. Each subsequent profile is reached with `.transformed(offset=( $\Delta x, \Delta y, \Delta z$ ), rotate=( $r_x, r_y, r_z$ ))`, where offsets follow a 2-D spline spine (same archetypes as sweep) and rotation angles are drawn from the discrete set $\{0, 10, 20, 30, 45, 60, 90\}^\circ$. Profile scale also varies per section. `.loft()` is called after all profiles are drawn.

Per-primitive fillet/chamfer. With 50% probability, each primitive solid is immediately followed by `solid_N = solid_N.edges().fillet(r)` or `.chamfer(d)`, $r, d \in [0.1, 0.9]$, applied as part of the primitive's own code block before it enters the Boolean tree.

H.4 SolidFactory: Bounding-Box-Driven Dispatch

SolidFactory wraps CADFeatureTreeGenerator and exposes a uniform bounding-box interface: every public method accepts $(c_x, c_y, c_z, s_x, s_y, s_z)$ (centre + extents) and returns a validated CadQuery solid together with the code string that built it. The internal flow is:

1. Convert centre/extents to a bbox dict.
2. Call the appropriate `generate_*` method. The generator selects a face, scales the sketch, and returns a feature dict.
3. Pass the feature dict to `generate_cadquery_code()`, which produces a Python string.
4. `exec()` the string and extract `final_result`.
5. Validate: single body, volume > 0.01, B-Rep valid (`BRepCheck_Analyzer.IsValid()`).
6. Strip the import header and `final_result` line; store the remaining code in `self._last_code` and the variable name in `self._last_var`.

Fallback hierarchy on failure: sketch-based extrude → cylinder → axis-aligned box.

H.5 ModelBuilderV2: Composing Primitives into Parts

ModelBuilderV2 is a stateful accumulator maintaining: `self.solid` (the current CadQuery solid), `self._code_lines` (a growing list of code-string fragments), and `self._model_var` (the Python variable name of the current combined solid).

At each step an operation is attempted in memory. If it produces a valid single-body solid, the operation's code fragment is appended and the solid updated; otherwise the step is silently skipped, making the builder failure-tolerant.

Base creation. `create_base()` calls SolidFactory with one of the four primitive types (or all four in random order until one succeeds) to create the initial solid, centred at the origin with depth extending along +Z. `create_nested_base()` builds a solid from an *outer* polygon profile with one or more *inner* cutout profiles (simulating perforated cross-sections such as frames or flanged tubes).

Union bosses. `add_boss()` / `add_boss_ex()` place a new primitive on a named face (top, bottom, front, back, left, right). `_face_region()` maps fractional width/height/depth parameters and a lateral offset (u, v) into a 3-D bbox, with an outward overlap to anchor the boss to the face surface. The primitive is generated inside that box and boolean-unions:

```
result = result.union(solid_N)
```

`add_step()` / `add_multi_step()` place wide, shallow bosses covering progressively smaller face fractions to build stepped profiles.

Intersection constraints for union bosses. All boss placement functions pass an `overlap_frac` parameter to `_face_region()`, which pushes the boss bbox inward by that fraction of the face depth so the boss is geometrically anchored to the host solid rather than merely touching it. The overlap fraction is sampled as $[0.05, 0.20]$ for regular bosses and $[0.05, 0.15]$ for steps. This is a placement-level geometric constraint (bbox overlap), not a volumetric ratio check.

Boolean cuts. `add_pocket()` and `add_through_cut()` compute an inward face region, generate a tool (90% database-sketch extrude, 5% cylinder, 5% rect box), then apply a volumetric intersection constraint before committing: `has_sufficient_intersection()` computes

$$\frac{V(\text{tool} \cap \text{solid})}{V(\text{tool})} \geq 0.20$$

and rejects the cut if the ratio falls below 20%, preventing tools that barely clip the surface.

```
result = result.cut(solid_N)
```

Through-cuts use a tool depth of $1.2\times$ the solid's full depth to guarantee complete penetration. `add_slot()` produces a narrow elongated cut (aspect ratio $\sim 6:1$). `add_nested_cut()` subtracts a nested outer-with-inner profile tool.

Spatial patterns. All three pattern types emit for-loop code.

Linear pattern — equally spaced tools along one face axis:

```
for i in range(N):
    pos = base + step * (i + 0.5)
    plane_i = cq.Plane(origin=(pos, cy, z0), ...)
    sketch_i = cq.Workplane(plane_i)
    sketch_i = sketch_i.circle(r)          # or polygon
    pat_tool = sketch_i.extrude(h)
    result = result.cut(pat_tool)         # or .union()
```

Circular pattern — N equally-spaced tools on a circle of radius ρ centred at (c_u, c_v) :

```
for i in range(N):
    angle = phi0 + 2*math.pi*i/N
    fu = cu + rho*math.cos(angle)
    fv = cv + rho*math.sin(angle)
    ...
```

Grid pattern — doubly-nested loop over rows $\times$ cols:

```
for ir in range(rows):
    for ic in range(cols):
        pu = bu + su*(ic+0.5)
        pv = bv + sv*(ir+0.5)
    ...
```

Each pattern commits only if ≥ 2 individual tool applications succeed.

Mirror. `add_mirror()` reflects the solid across XY, XZ, or YZ and unions the reflection. With 50% probability an off-centre mirror plane is used via a `basePointVector`.

Shell. `add_shell()` removes a selected face and hollows the solid to a specified wall thickness:

```
result = result.faces('>Z').shell(-t)
```

Intermediate fillet/chamfer. After union and cut operations, a 25–50% probability inline fillet or chamfer is applied to the sub-solid before the next operation, simulating mid-process edge-breaking in machining. Radius/distance is 1–4% of the current solid’s minimum dimension.

Final fillet/chamfer. `apply_fillet()` and `apply_chamfer()` are called at the end of a strategy, with radius/distance drawn as 2–20% of the minimum bounding-box dimension. On failure a smaller value (50%, 25%, or 10% of original) is tried automatically.

H.6 Code Assembly and Post-Processing

After construction, `get_code()` assembles the final self-contained Python script as follows:

1. Prepend `import cadquery as cq; import math.`
2. Join all `_code_lines` fragments in sequence.
3. Append `final_result = result` and `cq.exporters.export(final_result, 'model.step')`.
4. **Variable renumbering** (`_renumber_vars`): `plane_N`, `sketch_N`, `solid_N`, `loft_N`, `profile_N`, `path_N` indices are made contiguous. Gaps arise because the factory counter increments even on failed attempts.
5. **Variable substitution** (`_substitute_result_var`): `solid_0` is renamed to `result`; intermediate bare `result` assignments are removed.
6. **Float rounding** (optional, `-round-floats`): every float literal is rounded to three decimal places via regex substitution, reducing token count for language-model training.

7. **Collinear point removal** (`_remove_collinear_points`): any `lineTo` point that lies between its predecessor and successor on the same line, or duplicates either endpoint, is eliminated. Chained `.moveTo().lineTo()` calls are first split onto separate lines so that rounding-induced duplicates are also caught.

H.7 Strategy Library

Ninety-four named strategy functions orchestrate the builder, organised into seven tiers by complexity and intent:

- Tier 1 — Single-primitive (1–4).** One primitive type forms the entire body (`strategy_extruded_block`, `strategy_revolved_body`, `strategy_lofted_body`, `strategy_swept_body`).
- Tier 2 — Mid-level composition (5–20).** A primary primitive receives 2–6 additional features (bosses, pockets, patterns, mirror, shell). Examples: `strategy_symmetric_extruded` (`extrude` → features on one half → mirror), `strategy_lofted_enclosure` (`loft` → shell → mounting bosses).
- Tier 3 — Higher-level composition (21–30).** 10–18-feature sequences mimicking specific mechanical part families.
- Tier 4 — Union-focused (31–40).** Emphasise additive material: multi-scale boss clusters, tall fin arrays, nested cylinder rings, pancake stacks, hub-and-arm assemblies.
- Tier 5 — Mega-composite (41–48).** Combine two or more mid-level sub-strategies within one builder instance, yielding 12–25 features.
- Tier 6 — Long-sequence (49–72).** Detail-pass and nested-sketch strategies targeting 20–50+ features: surface nubs, micro-pockets, screw-boss panels, multi-hole plates, nested-sketch towers.
- Tier 7 — Operation-subset boost (73–94).** Each strategy restricts itself to underrepresented operations to rebalance the operation-type distribution: revolve-only, loft-only, sweep-only, cut-dominant, shell+fillet+chamfer focus, circular-pattern-only, mirror-heavy, sweep/loft alternating, grid/slot textured, revolve-cut lathe, circular-union crown, linear-union array.

H.8 Parallel Generation and Double Validation

Worker architecture. Generation is parallelised across W workers (default: all CPU cores). Each worker is an isolated subprocess spawned via `subprocess.Popen` rather than `multiprocessing.Process`, because forking from a `ThreadPoolExecutor` thread is unsafe with OpenCASCADE’s global C++ state. A per-model timeout (default: 120 s) sends `SIGKILL` to workers that hang inside the kernel.

Double validation. Each candidate model undergoes two independent validity checks:

1. **In-memory validation.** The builder solid must satisfy: (a) exactly one connected body (`count_solids=1`), (b) `volume > 0.01`, (c) `BRepCheck_Analyzer.IsValid()` returns `True`.
2. **Script re-execution validation.** The generated `.py` script is independently `exec()`-ed in a fresh namespace; the resulting solid must pass the same three checks. The STEP file is exported from *this* re-executed solid, guaranteeing that the stored script and the stored geometry are identical.

File management. Successful models are stored in chunk subdirectories of 10 000 files each (`0000/`, `0001/`, ...). Partial files from failed workers are cleaned up before committing. Completed runs write `metadata.json` (strategy, random seed, geometric scale, full feature history with parameters, axis-aligned bounding box, volume) and `strategy_stats.json` (per-strategy success/invalid/timeout/error counts).

H.9 Dataset Characteristics

Artifact types per model.

- `model_XXXXX.py`: self-contained CadQuery script (ground-truth program).
- `model_XXXXX.step`: STEP B-Rep geometry (exported from the re-executed script).
- `model_XXXXX.stl`: triangulated mesh (optional).

Program characteristics. Scripts consist of sequences of CadQuery API calls with explicit workplane definitions (`cq.Plane`), sketch primitives (`moveTo`, `lineTo`, `threePointArc`, `circle`, `spline`, `close`), volumetric operations (`extrude`, `revolve`, `loft`, `sweep`), Boolean combinators (`union`, `cut`), topology operations (`mirror`, `shell`), finishing operations (`fillet`, `chamfer`), and for-loop spatial patterns. Float literals are rounded to three decimal places.

I Policy Architecture and Training Details

I.1 Inputs

At each step, the model receives:

- **Target geometry m :** 1024-point cloud sampled from the input mesh.
- **Intermediate geometry S_{t-1} :** 1024-point cloud from the current reconstruction.
- **Geometry metadata:** bounding box center and scale of S_{t-1} in the target frame.
- **Program context:** the partial CAD program encoded as text tokens.

I.2 Point Cloud Encoding

Both target and intermediate point clouds are encoded using a Point-MAE encoder:

- Farthest Point Sampling: $1024 \rightarrow 64$ groups
- KNN grouping ($k = 32$)
- PointNet-based per-group encoder
- Transformer encoder (12 layers, partial fine-tuning)

This produces 64 tokens per point cloud, which are projected to the LLM embedding space.

I.3 Geometry Conditioning

A lightweight MLP encodes geometric metadata (center + scale) into a token that preserves global pose information lost during normalization.

I.4 Multimodal Token Fusion

Geometric tokens are injected into the language model by replacing reserved placeholder tokens in the input sequence. This yields a unified embedding stream combining:

- target geometry tokens
- intermediate geometry tokens
- geometric metadata
- CAD program text

I.5 Language Model

We use Qwen2.5-Coder-1.5B as the backbone. The model autoregressively predicts the next construction action as CadQuery code, including continuous parameters represented as digit sequences. Prompt: “Generate CAD operation $\{\text{op}_{i+1}\}$ for this shape:”

I.6 Decoding and Execution

At inference time:

- The model generates one construction action at a time
- The action is executed to produce updated geometry
- The resulting geometry is re-encoded for the next step

I.7 Training Setup

- Cross-entropy loss over operation tokens
- Full fine-tuning of the language model

- Separate learning rates for point cloud encoder (0.00005) and LLM (0.0002)
- 1000 learning-rate warmup steps followed by cosine decay
- Training on $\sim 12.5\text{M}$ state-action transitions
- Training performed on 8 NVIDIA H100 GPUs for 12 hours, batch size of 32, for 100k steps

I.8 Additional StepCAD Details

I.8.1 Problem Formulation

Let $m \in \mathcal{M}$ denote an input mesh, and let $Q_m \in \mathbb{R}^{N \times 3}$ denote the corresponding sampled point cloud. A CAD program is represented as

$$p = (a_1, \dots, a_T),$$

where $a_t \in \mathcal{A}$ are executable construction actions and $S(p)$ is the resulting geometry. The reconstruction objective is:

$$p^* = \arg \max_{p \in \mathcal{P}} \text{IoU}(S(p), m),$$

where $\mathcal{P}$ denotes the space of valid CAD programs.

Direct optimization over $\mathcal{P}$ is combinatorial and highly non-convex due to the discrete, sequential, and structured nature of CAD construction programs.

I.8.2 Autoregressive Policy Formulation

We model CAD generation as a sequential decision process. Let Q_t denote the point cloud sampled from the intermediate geometry S_t . At each step, the policy predicts:

$$\pi_\theta(a_t \mid Q_{t-1}, Q_m, a_{<t}),$$

which induces a distribution over complete programs:

$$\pi_\theta(p) = \prod_{t=1}^T \pi_\theta(a_t \mid Q_{t-1}, Q_m, a_{<t}).$$

Target and intermediate geometries are encoded using a shared Point-MAE encoder:

$$z_m = f_\phi(Q_m), \quad z_{t-1} = f_\phi(Q_{t-1}),$$

which are injected into the transformer as conditioning tokens.

The policy is instantiated using Qwen2.5-Coder-1.5B operating jointly over geometry embeddings and autoregressive program tokens. The next construction action is decoded as:

$$a_t = \arg \max_a \pi_\theta(a \mid z_m, z_{t-1}, a_{<t}),$$

or sampled from the predictive distribution during inference-time sampling.

I.8.3 Dense State-Action Supervision

The policy is trained using teacher forcing over dense state-action transitions extracted from ARCADE-1.5M:

$$\mathcal{L}(\theta) = -\mathbb{E}_{(Q_{t-1}, Q_m, a_t)} [\log \pi_\theta(a_t \mid Q_{t-1}, Q_m, a_{<t})].$$

Unlike supervision only on final programs, ARCADE-1.5M provides intermediate geometry after every solid-operation action, yielding dense trajectory-level supervision over approximately 12.5M construction states. This improves long-horizon robustness and geometric consistency.

I.8.4 Closed-Loop Rollout Procedure

At inference time, generation alternates between autoregressive prediction and CAD execution. Starting from an empty geometry $S_0 = \emptyset$, at each step:

1. The current intermediate geometry Q_{t-1} and target geometry Q_m are encoded.
2. The next construction action is sampled from

$$a_t \sim \pi_\theta(\cdot \mid Q_{t-1}, Q_m, a_{<t}),$$

using inference-time sampling.

3. The updated program is executed using CadQuery to obtain the next geometry:

$$S_t = \text{Exec}(a_1, \dots, a_t).$$

4. A new point cloud Q_t is sampled from S_t and fed back into the policy.
5. Reconstruction quality is evaluated using:

$$r_t = \text{IoU}(S_t, m).$$

Generation terminates once reconstruction quality plateaus (IoU improvement below 0.001), an end-of-program token is emitted, or $T_{\max} = 12$ construction actions have been emitted. This is an action cap, not a cap of 12 sketch/solid/Boolean operations or 12 API calls. ARCADE-1.5M averages 40.3 counted operations but only 8.5 solid operations per program, so programs containing more than 40 counted operations may fit within 12 actions. This does not guarantee recovery of shapes whose required construction exceeds the action budget.

This closed-loop execution is important because the policy conditions on the geometry actually produced so far rather than on teacher-forced intermediate states.

I.8.5 Motivation for Program Refinement

The learned policy efficiently captures the global structure of target shapes. However, due to the autoregressive and discretized nature of generation, small structural or parametric errors can accumulate over long construction sequences, motivating the geometry-aware refinement stage described in the main paper.

Algorithm 1 Geometry-aware tree search for CAD program refinement.

Require: Target mesh m , seed program $\hat{p} = (a_1, \dots, a_T)$, beam width K , local edits $\{e_1, \dots, e_M\}$

```
1:  $\mathcal{B}_0 \leftarrow \{\hat{p}\}$ 
2: for  $t = 1$  to  $T$  do
3:    $\mathcal{C} \leftarrow \emptyset$ 
4:   for  $p \in \mathcal{B}_{t-1}$  do
5:     for  $j = 1$  to  $M$  do
6:        $p' \leftarrow e_j(p, t)$ 
7:       Execute  $p'$  with CadQuery
8:        $s(p') \leftarrow \text{IoU}(S(p'), m)$ 
9:        $\mathcal{C} \leftarrow \mathcal{C} \cup \{(p', s(p'))\}$ 
10:    end for
11:  end for
12:   $\mathcal{B}_t \leftarrow$  top- $K$  candidates in  $\mathcal{C}$  ranked by  $s(\cdot)$ 
13: end for
14: return  $\arg \max_{p \in \mathcal{B}_T} \text{IoU}(S(p), m)$ 
```

J Geometry-Aware Program Optimization Details

J.1 Beam Search Procedure

Given a seed program $\hat{p} = (a_1, \dots, a_T)$ sampled from the policy π_θ , we perform local optimization in program space:

$$p^* = \arg \max_{p \in \mathcal{N}(\hat{p})} \text{IoU}(S(p), m),$$

where $S(p)$ denotes the geometry produced by executing program p and $\mathcal{N}(\hat{p})$ is a neighborhood induced by local program edits.

Searching all edits jointly is exponential in the sequence length T . Instead, we exploit the sequential structure of CAD programs and perform a geometry-aware beam search over operations in execution order. At depth t , each candidate in the current beam is expanded by applying local edits to operation a_t , yielding:

$$\mathcal{B}_t = \text{TopK}_K \{p' \in \text{Expand}(p, t) : p \in \mathcal{B}_{t-1}\},$$

where candidates are ranked by:

$$\text{score}(p) = \text{IoU}(S(p), m).$$

We use beam width $K = 2$ and $M = 4$ local edits per expansion. At depth $t = 1$, the initial program expands into four candidates and the best two are retained. At subsequent depths, each surviving candidate produces four neighbors, resulting in eight evaluated candidates per depth.

The final refined program is selected as:

$$p^* = \arg \max_{p \in \mathcal{B}_T} \text{IoU}(S(p), m).$$

J.2 Continuous Parameter Refinement

The continuous refinement operator optimizes the scalar parameters of the current operation without modifying program structure. This includes extrusion depths, radii, sketch coordinates, offsets, and angles.

For an operation parameter vector

$$v \in \mathbb{R}^d,$$

we define a bounded local search region:

$$[v_i - r_i, v_i + r_i], \quad r_i = \min(\max(0.20|v_i|, 0.005), 0.30).$$

Optimization is performed using Bayesian Optimization with a Gaussian-Process surrogate and Expected-Improvement acquisition. We initialize the search with 6 random samples followed by 10

BO iterations using batches of 4 candidates each. For fillet radii and chamfer lengths, we instead use the fixed interval $[0.01, 0.5]$ in normalized coordinate units.

The refined parameters are accepted only if the resulting geometry improves IoU relative to the original operation.

J.3 Sketch Extraction and Replacement

The sketch modification operator replaces the sketch associated with the current operation using geometry extracted directly from the target mesh.

Given the operation working plane $(O, \vec{n}, \vec{x})$, we intersect the target mesh with the plane using `trimesh.section`. The resulting curves are projected into the local 2D coordinate system $(\vec{x}, \vec{y})$ and converted into closed polygons. The largest valid closed loop is selected.

The polygon is simplified using the Ramer–Douglas–Peucker algorithm, after which a primitive-fitting procedure selects among:

- full circles,
- line-and-arc decompositions,
- raw polygonal approximations.

The reconstructed sketch is then serialized back into CadQuery sketch commands:

```
Workplane(...).moveTo(...).lineTo(...).threePointArc(...).close()
```

The original operation sketch is replaced while preserving the operation type and extrusion depth.

This edit is skipped if the plane does not intersect the target mesh or only open curves are produced.

J.4 Operation Skipping

The operation-skipping edit removes the current operation entirely:

$$p' = p \setminus a_t.$$

The modified program is executed and accepted only if:

$$\text{IoU}(S(p'), m) > \text{IoU}(S(p), m).$$

This edit is particularly effective for removing redundant operations, degenerate sketches, or erroneous boolean cuts introduced by the autoregressive policy.

J.5 Complement-Cut Optimization

The complement-cut operator removes overshoot regions by generating cut-extrudes from the boolean difference between the generated geometry and the target mesh.

Let:

$$G = S(p), \quad I = m.$$

We compute the overshoot region:

$$D = G \setminus I$$

using `manifold3d` boolean operations.

Planar faces of D are clustered by normal direction and plane offset using:

- angular tolerance: 2° ,
- offset tolerance: 10^{-3} .

Clusters with total area below 0.0025 are discarded.

For each planar cluster, boundary loops are projected into the local 2D plane and converted into sketches using the same primitive-fitting procedure as sketch replacement. Candidate cut-extrudes are generated by grid-searching extrusion depths:

$$\{0.05, 0.10, 0.15, 0.20, 0.30, 0.40, 0.60, 0.80, 1.20, 2.00\}.$$

The best cut operation is appended to the program using:

`combine='cut'.`

We allow up to two complement-cut rounds, since removing one overshoot region can expose additional dominant errors.

K Additional Evaluation: Fidelity, Search Transfer, and Compactness

K.1 High-IoU Reconstruction Rates

We report the percentage of evaluated CADBench samples reaching each IoU threshold. High-threshold success measures volumetric fidelity without requiring program correspondence. It does not imply that all dimensions or small features are correct, because global volume overlap can be insensitive to localized errors.

Table 9: **High-IoU success rates on CADBench (%)**. Thresholds assess volumetric reconstruction fidelity, not direct parameter accuracy.

Method	≥ 0.80	≥ 0.90	≥ 0.95	≥ 0.99
CAD-Recode	43.8	30.6	19.5	3.9
Cadrille	44.1	30.2	18.7	3.4
CADEvolve	45.9	32.3	21.7	5.6
CADReasoner	49.4	38.4	27.2	7.4
StepCAD	71.1	55.3	39.9	15.3

K.2 Transfer to Independently Generated Programs

The refinement stage uses the target mesh and an executable seed program, with no policy weights or learned policy features. On 1,200 CADBench samples, applying it to CAD-Recode output increases mean IoU by 0.148. This demonstrates transfer to another generator’s compatible programs; it does not establish support for arbitrary CAD languages or all program structures. The subset results are separate from the full-benchmark results in Table 9.

Table 10: **Search applied to CAD-Recode programs** on the same 1,200-sample CADBench subset, without retraining.

Metric	CAD-Recode seed	With StepCAD search
Mean IoU	0.597	0.745
IoU ≥ 0.80 (%)	45.8	63.4
IoU ≥ 0.90 (%)	29.2	45.8
IoU ≥ 0.95 (%)	17.1	32.6

K.3 Entity-Level Geometric Accuracy

On CADBench datasets with reference B-reps, we additionally evaluate vertex, edge, and face accuracy using the definitions and tolerance of Pointer-CAD v2 [40]. StepCAD has the highest reported accuracy at all three levels (Table 11). These metrics provide information not captured by global IoU. Their interpretation depends on the matching protocol and reference decomposition: different valid

histories can yield similar geometry with different surface representations or subdivisions. This limits their use as a unique measure of construction-history correctness, but does not make entity-level geometry uninformative.

Table 11: **Entity-level accuracy** on CADBench datasets with reference B-reps. Values are the accuracy scores reported in the additional evaluation.

Method	Vertex $\uparrow$	Edge $\uparrow$	Face $\uparrow$
CAD-Recode	11.4	6.3	4.1
CADEvolve	15.7	9.7	6.4
StepCAD	20.8	12.6	8.1

K.4 Program Size and Reconstruction Quality

Table 12 reports mean total counted operations together with IoU. The policy alone produces the shortest programs among the listed methods while exceeding their reported IoU. Search increases mean size from 16.2 to 31.7 operations as IoU improves from approximately 0.68 to 0.82. Thus, refinement trades program length for fidelity; it does not uniformly simplify the construction history. Operation counts characterize compactness but do not establish constraint preservation, semantic correctness, or successful downstream editing.

Table 12: **Program size and fidelity on CADBench.** Counts are total operations rather than policy actions; IoU is reported at the precision provided by this analysis.

Method	Mean operations	IoU $\uparrow$
II-StepCAD	16.2	0.68
CADEvolve	19.9	0.57
Cadrille	23.5	0.58
CAD-Recode	25.5	0.57
StepCAD	31.7	0.82

K.5 Conciseness Relative to Reference Programs

Following the motivation of Fusion 360 Gallery [20], we compare recovered sequence length with reference sequence length on DeepCAD. For recovered and reference sequences A and $\hat{A}$, respectively, the ratio is $|A|/|\hat{A}|$, using consistent counting units. Table 13 reports the supplied analysis stratified by B-rep face-count complexity. StepCAD has an overall ratio of 0.91 and IoU of 0.89. A smaller ratio is not necessarily better: omitted geometry can shorten a program, so conciseness must be interpreted jointly with reconstruction accuracy. These results are not restricted to exact reconstructions.

Table 13: **Conciseness on DeepCAD.** Recovered/reference sequence-length ratios and overall IoU. Lower ratios alone do not imply better reconstruction or editability.

Method	Easy	Medium	Hard	Overall	IoU
CAD-Recode	0.99	0.99	0.71	0.90	0.79
CADEvolve	1.02	0.82	0.53	0.79	0.74
II-StepCAD	0.95	0.87	0.57	0.80	0.82
StepCAD	1.05	0.94	0.73	0.91	0.89

Assets and licenses

We evaluate on CADBench, which aggregates data from publicly available datasets (e.g., ABC, Fusion 360, Objaverse). We follow the respective licenses and terms of use of the original sources. We also use pretrained models (Qwen2.5-Coder, Point-MAE) and CadQuery (Apache 2.0), in accordance with their licenses.

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